# Recorded metadata predicts the phenotype label in eight of nine public single-cell cohort comparisons

**Short title:** Study metadata and single-cell patient classification

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## Abstract

Patient-level classifiers reduce thousands of single cells to one score per donor and routinely report cross-validated AUC above 0.95. Such a score is a statement about disease only if the variables recorded about how a donor was recruited, processed and sequenced do not themselves predict which group that donor is in. We measured that association in nine phenotype contrasts from five public blood datasets (809 donors), covering lupus, COVID-19, influenza, cytomegalovirus serostatus and sepsis versus COVID-19. Recorded metadata were split into collection, demographic and sample-quality variables, and each group's out-of-fold AUC judged against a permutation null built from that comparison's own metadata matrix. Two permutation tests were then applied to the expression classifier: one permuting the phenotype label freely, one permuting it only within strata sharing a collection configuration.

Metadata alone predicted the phenotype at out-of-fold AUC 0.404 to 1.000, significantly in eight of nine comparisons in all five split seeds and after Benjamini-Hochberg correction within each seed; collection variables alone were significant in four of the seven comparisons recording any. The exception was a cytomegalovirus comparison specified in advance as a negative control: its metadata model stayed near chance while its expression classifier reached AUC 0.894 to 0.933. At the other extreme, a 21-donor influenza comparison had no collection stratum holding both labels, which makes the stratified permutation uninformative rather than non-significant. Between these ends the variable groups separated cohorts a single number would have equated: two COVID-19 cohorts with similar overall metadata AUC were driven by sample quality in one and recruiting site in the other. Adding expression to the metadata raised AUC by 0.087 to 0.354 in seven comparisons and by nothing measurable in the other two.

These analyses measure association with recorded variables. They do not establish that a classifier's discrimination is biological.

**Keywords:** patient-level classification; single-cell RNA sequencing; study metadata; confounding; permutation tests; external validation; systemic lupus erythematosus; COVID-19

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## Author summary

Single-cell sequencing of a blood sample can be turned into a diagnostic score, and published scores often separate patients from healthy donors almost perfectly. Before trusting one, a clinician needs to know whether it reads the disease or reads the study. In most public datasets patients and healthy donors were not recruited the same way: they may come from different hospitals, years or laboratory runs, and a model can score well simply by recognising where a sample came from.

We measured how large this problem is across nine comparisons from five public datasets - lupus, COVID-19, influenza and cytomegalovirus infection - by asking how well the recorded study details alone predict which group a donor is in. In eight of the nine they predict it well above chance. In the ninth, chosen in advance as a negative control, they predict nothing while the expression classifier still separates the groups. Which details carry the prediction differs between cohorts that look equally affected overall.

The usual test of these models shuffles the patient labels, which destroys the pattern that needs examining. Shuffling labels only among donors collected the same way asks the complementary question, and in one comparison the two disagree.

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## 1. Introduction

Single-cell RNA sequencing supports patient-level classification by capturing differences in immune-cell composition and molecular state, and published classifiers routinely separate patients from controls at cross-validated AUC above 0.95. For such a result to inform clinical research, the discrimination has to extend beyond the recruitment and processing conditions of the development cohort. That requirement is hard to assess when the phenotype label is associated with collection site, sequencing batch or sample quality.

A growing family of methods competes to produce that one number per patient: pseudobulk aggregation, phenotype-oriented summaries, prototype networks, sparse gene panels, multicellular factor models, graph representations, and methods that first identify the disease-relevant cell population [6-11,32-35]. Frozen single-cell foundation models offer a convenient route - encode each cell once, average, train a small classifier on donors - though benchmarks report that such embeddings can retain batch structure and can lose to simple expression baselines [13,14]. Our concern is not which of these wins. It is that the comparison between them is made on cohorts where the winner may be decided by something other than biology.

That question is hard to answer in public data because cases and controls are usually not collected the same way. They may be recruited from different clinics, sequenced in different years, run on different chemistries, or processed in different batches. When that happens, a model can reach a high accuracy by learning the collection process. Batch-label confounding has been known to bias cross-validation for a decade [16], and a fully confounded single-cell design is not identifiable [17]. Adjusting for technical covariates does not fix it: regression removes the part of the data that design can predict, whether that part is technical, biological, or both [18,19]. Work in predictive modelling has shown the same thing from three directions - adjustment must be done inside training folds, confound regression can leak information to nonlinear learners, and residual confounding survives apparently successful correction [27-29]. Benchmark-side work raises the matching concern about how the datasets themselves are built [12,15].

Methods to quantify confounding and to test conditional association already exist, and we use them: the permutation approach of Chaibub Neto and colleagues [30], the model-level confounding statistics of Spisak [29] and the Confounding Index of Ferrari and colleagues [25] all address the same estimand from different directions. What has not been established is what those methods return when they are applied across the public single-cell cohorts on which patient-level methods are benchmarked, where the acquisition structure differs from cohort to cohort and much of the relevant metadata is missing or partial. We therefore ask four questions of each cohort. How strongly does the recorded metadata predict the phenotype label? Which groups of variables carry that prediction? How does the answer compare with what the same metadata matrix would produce under a permuted label? And when is the question answerable at all?

A word on "recorded metadata", because the distinction runs through the whole paper. The released tables mix three kinds of variable. Collection facts are study design in the strict sense. Demographics are case mix, and may also be genuine risk factors. Sample-quality summaries sit downstream of the assay, but can sit downstream of the disease too. Lumping them together and calling the result "study design" would overstate what a positive result means. We keep them apart, treat collection as primary, and report the headline count under both definitions.

We analyse nine phenotype contrasts from five public single-cell datasets and examine what metadata association does to donor-level validation. First, we separate collection variables, demographics and sample-quality measures, and identify which groups predict the phenotype. Second, we compare a free label permutation with one confined to collection strata, to ask whether discrimination exceeds the association those strata retain. Third, we use simulations and two detailed lupus analyses to examine why residualisation, within-stratum restriction and external validation can support different readings of the same cohort.

Three results follow. The association between recorded metadata and the phenotype label is graded and measurable, and varies across routinely reused cohorts by more than half the usable AUC scale. It can be traced to an interpretable group of variables, and cohorts with near-identical overall metadata AUC are driven by different groups. And the permutation test normally used to validate these pipelines states a different null from the one at issue; we give a comparison where the two tests point in different directions, and we measure what the collection-stratified test does and does not cover.

Section 8 states the scope of each quantity and what the approach cannot see.

# 2. Study framework and evaluation measures

## 2.1. The setting

Figure 1 sets out the situation in the notation of graphical causal models [36,37]. A recruitment process **R** assigns each donor a place in the recorded design **D** - site, batch, calendar period, chemistry, sample-quality profile - and is also associated with the diagnosis **Y**, because cases and controls are usually found through different routes. The measured cells **X** are affected by **D** through technical effects and by **Y** through biological ones. A classifier maps X to a prediction $\hat{Y}$.

Two further nodes matter in practice.

**C** stands for covariates that are consequences of the disease: severity, WHO score, comorbidity, medication, time since symptom onset. These sit on the path $Y \to C \to X$. They are not collection variables. Treating them as metadata variables converts "the disease is predictable" into "the design is predictable", and Section 4.1 quantifies how badly.

**U** stands for collection structure that was never recorded. Nothing in this paper can see U. That is a limit of the data rather than of the method, and Section 8 says so.

The graph was specified before the analysis and is released in machine-checkable `dagitty` syntax [38].

One thing the arrow between D and Y is not. We draw the association as induced by R rather than as a directed edge. Both directions occur in practice. A patient may be recruited at a referral centre because they are a patient, and a centre's case mix may determine which diagnoses appear in a batch. These checks measure the association. They do not tell you which way it runs.

## 2.2. The primary measure, and the variance summary reported beside it

**The primary measure is the metadata-only AUC.** For each comparison we fit a classifier on the columns of the recorded metadata matrix **D** and nothing else, and report its out-of-fold AUC. Significance comes from permuting the phenotype label 200 times and pushing each permuted label through the identical fixed-hyperparameter pipeline, so the observed statistic and its null are the same quantity computed the same way. Which columns are in D is stated every time it is used, and three names are kept distinct throughout: the **metadata-only AUC** uses every recorded variable, the **collection-only AUC** uses the collection block alone, and the **demographic-only** and **sample-quality-only** AUCs use those blocks. The same permutation test is run for each group, and a nonlinear arm is run alongside the linear one so that a low value is not an artefact of assuming linearity.

**A linear variance summary is reported beside it.** Let $y$ be the centred phenotype vector. Write

$$V\_D = 1 - R^2(y \sim D).$$

This is the share of phenotype variation that the recorded metadata cannot explain in a linear model. It is a residual variance, not a Shannon or Fisher information: it is linear, and it depends on how D is coded. Two things make it readable.

*Cross-fitting.* An in-sample $R^2$ inflates with the ratio of metadata columns to donors, so V_D is also computed out-of-fold with five-fold cross-fitting. The correction is not cosmetic. The CMV comparison declares 89 metadata columns for 108 donors, so its in-sample $R^2$ is inflated to 0.532 and its in-sample V_D falls to 0.468, which would rank it the third-most affected of the nine. Its matched in-sample null sits at 0.496: the observed in-sample V_D is not significantly lower than its own permutation reference distribution.

*A null built from the same matrix.* The phenotype is permuted 200 times and each permuted label is pushed through the same five folds, giving a one-sided p-value, p(V_D), for that specific matrix at that specific donor count. A raw V_D cannot be compared across cohorts; its position inside its own null can. The in-sample pair is released alongside with its own in-sample null; p(V_D) refers throughout to the cross-fitted pair, and p(V_D, in-sample) to the other.

**Why the AUC and not V_D carries the test.** V_D is an *unpenalised* linear $R^2$, and cross-fitting an unpenalised fit loses power as the matrix gets wider. The Ren COVID-19 comparison shows this: its 26 one-hot columns on 182 donors leave the cross-fitted V_D at its ceiling, because the linear model explains nothing out of fold. A penalised logistic model on the very same matrix reaches AUC 0.81 to 0.85, and its own permutation null puts that at $p = 0.005$ in every seed. Reporting only p(V_D) there would say “no association” about a comparison in which the recorded metadata plainly predicts the phenotype. So the primary test is the permutation null for the metadata-only AUC; V_D and p(V_D) are the linear-variance summary, and where the two disagree we say so.

**What this does and does not cover.** Both quantities look at one arrow in Figure 1, the one between D and Y. Confounding of a classifier also requires that D affects X and that the classifier uses that part of X. A high metadata-only AUC is therefore necessary but not sufficient evidence that a reported expression AUC is metadata-driven. A low one does not rule out an association through unrecorded structure. Every use of these quantities below is subject to this.

### 2.3. Two permutation tests that ask different questions

The **free permutation** shuffles the phenotype label without restriction and refits the whole pipeline. Its null is that the label is exchangeable across all donors, so rejecting it is evidence of an association between expression and the label - of any origin. It is the test these pipelines are normally validated with, and we keep it, because a pipeline that cannot reject it is not measuring anything. What it does not do is separate the two origins: a free permutation destroys the metadata-label association along with the biology, so a small p-value is equally consistent with either. Nor does rejecting it certify that the pipeline is free of leakage. A label-dependent step taken *before* the permutation - a feature list carried in from a prior analysis, say - is present in the observed and the permuted runs alike, and this test cannot see it. Every representation step in the pipeline used here is refitted inside the training fold on every permutation (Section 9.6), which is the property that has to be arranged in the code; the permutation test does not establish it.

The **collection-stratified permutation** shuffles the phenotype label only *within* strata that share a collection configuration - batch, pool, site and, where it varies, sequencing chemistry - so that each stratum keeps its observed class counts and only the correspondence between labels and molecular profiles is randomised inside it. This is the restricted permutation scheme introduced by Chaibub Neto and colleagues [30] and used since in the confounding literature [25]; it is not new here, and what is new is the comparison of what the two tests return across cohorts with different acquisition structures. Its null is conditional exchangeability given those strata, so rejecting it is evidence of discrimination beyond what the recorded collection strata retain.

**What it conditions on, and what it does not.** The strata are built from collection variables only. The recorded metadata also contains sample-quality and demographic columns, and those are *not* conditioned on. So a significant result means the classifier scored above what these collection strata retain. It does not mean the classifier beat everything recorded, and it is not evidence that some part of the discrimination is free of association with the recorded metadata. If a sample-quality variable is still associated with the phenotype inside a stratum, and drives expression, then the conditional-independence null is false for a reason that has nothing to do with disease, and a classifier can reject it with no disease effect present at all. Section 3.4 measures how often that happens rather than assuming it does not. Conversely, a non-significant result is a failure to reject under this conditioning set, not a demonstration that the discrimination is entirely collection-driven.

Conditioning on the full metadata instead would require a conditional randomisation scheme for a matrix containing continuous columns, which is a different and harder problem; we state the limit rather than overreach past it.

The two tests are not defined on the same data. The collection-stratified test needs at least one stratum containing both classes. When every stratum is label-pure the within-stratum shuffle leaves all labels unchanged, the permutation group contains only the observed labelling, and the reference distribution is degenerate. We report the test as **uninformative** and its p-value as NA, because it cannot distinguish the observed result from a non-trivial reference; the free permutation is still defined in that situation and is still reported. Section 4.4 gives a comparison where exactly this happens.

### 2.4. Three verdicts: estimable, underpowered, and not answerable with these data

We separate two failure modes that are easy to run together and that call for different responses.

A comparison **cannot be answered with these data** when the quantity has no meaning for it: no collection stratum holds both a case and a control, so the collection-stratified permutation cannot be formed. These are dropped from the main ordering and reported separately.

A discrepancy between a tuned and a fixed-hyperparameter classifier is deliberately **not** used as a feasibility rule. Comparing two algorithms across a cut-point would not establish that a comparison is unanswerable, and a tuned model outperforming a fixed one is not the same kind of problem as a comparison with no donors collected under overlapping conditions. The fixed-hyperparameter pipeline is the inferential statistic everywhere, because it is what both permutation nulls are built from. The tuned AUC sits beside it as a description. Their difference is released as a column, so a reader can judge pipeline sensitivity directly instead of through a threshold.

A comparison is **underpowered** when the quantity is well defined but unstable: fewer than 40 donors, or a minority class below 15. These stay in the ordering, flagged. Deleting them would hide the instability a reader needs to see. The gates themselves are deterministic - they depend on donor counts and stratum structure, not on the seed - but what an underpowered comparison then reports is not: Section 4.4 gives one whose collection-stratified p ranges from 0.057 to 0.614 over five seeds, an order of magnitude of movement in a quantity that never crosses 0.05.

# Results

Sections 3 to 6 report results. Section 3 establishes how the checks behave on data with a known generating mechanism. Section 4 applies them to nine phenotype contrasts. Section 5 asks whether a classifier actually uses the structure the checks detect. Section 6 states what each check can and cannot support.

## 3. How the checks behave in simulation

A simulation with one binary design variable, an additive design effect and constant noise would not show whether these checks behave when those choices are relaxed. We therefore crossed seven data-generating regimes with seven levels of metadata-phenotype association and two outcome types, binary and continuous, each predicted and scored on its own scale: 19,600 simulated cohorts of 200 donors and 100 features (Figure 2). Nothing in the simulation is fitted to the real cohorts.

### 3.1. Calibration of the metadata-association statistic

When design carries no information about the diagnosis, the checks must not fire. They do not. The fraction of runs with p(V_D) $\leq$ 0.05 is 0.025 to 0.065 across the seven regimes, which brackets the nominal 0.05 (Figure 2A). It rises at $\rho = 0.25$ and reaches one by $\rho = 0.50$ in every regime.

This is the property the cross-fitted estimator and the per-cohort null were introduced to obtain; an in-sample estimator judged against an in-sample null does not have it.

### 3.2. Residualisation and restriction give different answers, in every regime

Two ways of handling design are in common use. Residualisation regresses the data on the metadata variables and analyses what is left. Restriction compares only donors collected under matching conditions. In the additive linear regime the two agree when design carries no diagnosis information and separate as it does (Figure 2B). At $\rho = 0$ the unadjusted, residualised, restricted and external AUCs are 0.809, 0.809, 0.787 and 0.817. At $\rho = 0.98$ they are 0.879, 0.538, 0.724 and 0.891. Residualisation has removed 0.34 AUC and restriction 0.16, on data where the biological effect never changed. At $\rho = 1$ restriction is not merely worse. It is undefined, because no stratum holds both labels.

The gap between the two opens in **every** regime (Figure 2C), so it is not an artefact of the linear setting.

The continuous-outcome arm is a genuine test of outcome type here, not a relabelling. Its cohorts are generated with a continuous latent outcome, predicted with ridge regression, and scored by concordance over pairs, which equals AUC when the outcome is binary and so puts both arms on one scale. The ordering survives: restriction beats residualisation at rho = 0.98 in all seven regimes for both outcome types. The magnitude does not. With a binary outcome the gap is 0.094 to 0.200; with a continuous one it is 0.013 to 0.038. So the qualitative conclusion does not depend on outcome type, and the quantitative one does - a simulation run only on continuous outcomes would have understated the problem by roughly a factor of five.

### 3.3. Effects of class prevalence and of target acquisition structure

**What a rare diagnosis does, and what it does not.** Two things travel together in an imbalanced cohort and are easy to confuse. One is prevalence itself. The other is that a collection boundary drawn at the middle of a population no longer lines up with a phenotype drawn from its top fifth, so the metadata cannot separate the labels even when the two are perfectly linked in the latent model. We separate them with two arms. In `imbalanced`, both thresholds sit at the 80th percentile, so prevalence is the only thing that changed. In `imbalanced_offset`, the diagnosis sits at the 80th percentile while the collection boundary stays at the median. The result is that prevalence alone changes nothing. With thresholds matched, the imbalanced arm tracks the balanced one almost exactly: metadata-only AUC still reaches 1.000 at rho = 1 and V_D still falls to 0.000. It is the offset arm that blunts the checks, with metadata-only AUC saturating at 0.880 and V_D flooring at 0.728. The reassurance a rare diagnosis appears to give is therefore not about how rare it is. It is about whether the collection boundaries line up with the diagnosis, and a cohort can have that problem at any case rate.

Two of our own comparisons have minority classes of 10 and 11 donors, so this matters for reading them.

**An external cohort collected the same way will confirm a metadata-driven model.** In the base regime the external AUC does not fall as confounding rises. It *rises*, from 0.817 at $\rho = 0$ to 0.904 at $\rho = 1$, tracking the internal AUC almost exactly (Figure 2D). Only when the target is collected under a different design mechanism does the external estimate come apart, flattening at 0.816-0.829 across the whole range while the internal estimate climbs to 0.899.

The practical reading is uncomfortable. External validation checks design confounding only to the extent that the second cohort's collection process is genuinely independent of the first. Being a different dataset does not establish that. Two cohorts assembled by similar consortia, on similar platforms, through similar recruitment routes, can agree with each other and both be wrong. This sharpens Check 5 in Section 6 and is, in our reading, the most consequential thing the simulation adds.

### 3.4. A non-disease pathway outside the conditioning set makes the stratified test reject

The collection-stratified permutation conditions on collection strata. The recorded metadata also holds sample-quality and demographic columns, which it does not condition on. If one of those is associated with the phenotype *inside* a stratum, and also drives expression, then the classifier can reject the stratified null with no disease effect present anywhere. This deserves a number rather than a caveat.

We simulated it directly. Cohorts of 200 donors are assigned to eight collection sites. A sample-quality variable QC predicts the phenotype within each site, at a strength we vary. Expression depends on the site and on QC, and on nothing else - the generating model contains no arrow from the phenotype to expression at all. Both permutation tests then run exactly as they run on the real cohorts (Figure S1, Table S1).

Be precise about what the rejection rate is for each . The generating model is QC $\rightarrow$ Y and QC $\rightarrow$ X, so conditioning on the collection strata leaves X and Y dependent: for $> 0$ the conditional-exchangeability null the stratified test states is **false**, and rejecting it is a correct rejection of a false null, not a type-I error. Only the $= 0$ row measures calibration. At $= 0$ both tests are calibrated: the collection-stratified permutation rejects in 19 of 300 runs, 6.3% (exact 95% CI 3.9 to 9.7; binomial $p = 0.29$ against a nominal 5%), and the free permutation in 21 of 300, 7.0% (4.4 to 10.5; $p = 0.11$). The classifier's mean AUC over those runs is 0.500.

Above $= 0$ the rate rises with the strength of the within-stratum association and the two tests move together, neither more protective than the other. At a mean within-stratum absolute correlation of 0.29 the stratified test rejects in 22.0% of runs and the free test in 22.7%; at 0.61 they reject in 95.3% and 95.7%. The classifier's own AUC rises from 0.500 to 0.668 over the same range, with no disease effect anywhere in the generating model.

That is the useful result, and it is a statement about interpretation rather than about error rates. A significant collection-stratified result is evidence that the phenotype is associated with expression beyond what the collection strata retain. It is **not** evidence that the association is disease biology, because a recorded variable outside the conditioning set can carry it. Where a sample-quality or demographic block is itself associated with the phenotype - which Table 1 reports for every comparison - that possibility has to stay open.

### 3.5. Residualisation loss without any metadata-phenotype association

Residualisation loss is often read as a measure of how contaminated a cohort was. It is not. The size of the loss that has nothing to do with contamination can be measured, and it turns out to be the same size as the loss observed in the cohort where nothing is detected.

We generated cohorts in which the metadata matrix is independent of the phenotype by construction, with a real biological effect calibrated so the unadjusted AUC lands near 0.90, and residualised on matrices of increasing width. The unadjusted and residualised arms are the same call to the same function with one argument added, so folds, feature filter, standardiser, PCA, classifier and scoring rule are shared and the difference between them is the adjustment alone (Section 9.9). A second condition reuses the exact level sizes of the real CMV batch-by-pool crossing, which is far more ragged than equal blocks: 46 levels for 108 donors, 17 of them holding one person.

Loss appears without any metadata-phenotype association, and it is large. Its size tracks how much of the sample the matrix can address rather than the column count alone. At one to eight columns residualisation costs almost nothing - mean loss 0.004 to 0.023, because removing a few directions leaves the biological one intact. Loss then rises steeply, peaking at 0.333 for 48 columns and 0.302 for 64, and falls back to 0.222 at 80 and 0.159 at 92, the width closest to the CMV comparison's 89: past a point, extra one-hot columns are increasingly redundant with the ones already there and ridge shrinks them together. Run-to-run spread is wide throughout; the central 95% of runs at 48 columns spans 0.224 to 0.460.

The ragged CMV-shaped layout is the one to read. It realises 45 columns from the observed pool sizes and costs **0.363** on average, central 95% 0.272 to 0.464 - more than balanced blocks of any width, and enough to contain the 0.326 loss the real CMV comparison shows (Section 5.3, Figure S2). The metadata-only AUC in these simulated cohorts stays at 0.48 to 0.51 throughout, confirming that the matrix carries no phenotype information to remove. Residualising expression on a wide, ragged matrix over 108 donors therefore costs about a third of an AUC point's separation whether or not there is anything to remove, and a loss of that size is not evidence that there was.

Two limits on how far this goes. Column count is not effective dimension: the real CMV collection block has 80 columns at centred rank 45, and equal blocks of 80 columns span a different subspace, which is why the ragged layout exists. And the simulation fixes one adjustment procedure, ridge at a fixed penalty on a training-fold standardised matrix; a differently penalised residualiser would trade loss against completeness differently.

# 4. Nine phenotype contrasts from five public datasets

## 4.1. Datasets, and what the recorded metadata contains

We assembled five public datasets [2,39-42] through the CELLxGENE Discover curation API and reduced each to one row per donor: log1p CPM pseudobulk expression, plus variables built from schema-guaranteed observation fields (`donor_id`, `disease`, `sex`, `development_stage`, `assay`, `tissue`) and cohort-specific collection variables recovered from the distributed object.

**These three groups are not the same kind of thing, and we do not treat them as one.**

- **Collection** (site, sub-study, sequencing chemistry, assay, batch, pool) is study design in the strict sense: it describes how the material was gathered and processed, and nothing about it is caused by the disease.
- **Demographic** (age, sex, ethnicity) is case mix. An imbalance here is a property of who was recruited, so it is a design fact; but age and sex are also genuine risk factors for most of these diagnoses, so an association with the label can be aetiological rather than procedural. The two readings cannot be separated from the data.
- **Sample quality** (cells per donor, mean UMI, genes per cell, mitochondrial fraction) is measured downstream of everything. It reflects the assay, but it can also reflect the disease, through cell composition, treatment or the state of the sample at collection. Section 4.1's mediator result shows exactly how much damage that can do.

Because of this, **collection is the primary definition** wherever a result is read as a statement about study design, and the other two groups are reported beside it as prespecified sensitivity analyses. The union of all three is what we call the *recorded metadata*, and every claim about the union is worded as such. Section 4.2 gives the headline count under both definitions, because they differ.

Covariates that are consequences of the disease are excluded from all three groups by an explicit list: severity, outcome, comorbidity, medication, symptom timing, diagnosis fields, stage, grade, WHO score. These are the C node of Figure 1.

The exact raw columns and expanded metadata columns entering every group, for every comparison, are generated directly from the model-matrix construction code and released as Table S2. A machine check confirms that the released inventory reproduces the width of every model matrix actually fitted (34 of 34 blocks) and that no diagnosis label or label proxy appears in any of them.

This exclusion is not a formality, and we quantified what it is worth. In a simulation where metadata and phenotype are completely unrelated, admitting a single disease-caused covariate to the metadata variables raises the metadata-only AUC from 0.488 to 0.793. It also flags the cohort as significantly affected in **100% of runs**, against 5.5% without it (Supplementary Figure S3, Supplementary Table S3). Residualising on the enlarged set then removes 0.12 AUC from a cohort in which nothing was confounded at all. A single mediator admitted by accident is enough to manufacture the entire finding.

Nine case-control comparisons were formed (Table 1): lupus versus healthy in GSE174188 (CD4 subset and all cells), COVID-19 versus healthy in three independent cohorts, influenza versus healthy and sepsis versus COVID-19 in COMBAT, and CMV-positive versus CMV-negative in the HIHA cohort. The five cohorts hold 809 distinct donors (261, 196, 120, 124 and 108). Comparisons drawn from the same cohort share donors, so the per-comparison counts in Table 1 do not sum to a donor total.

| Comparison | Donors | Minority | Metadata AUC | p | Collection AUC | p | Expression AUC | + over metadata | p free | p strat. | Verdict |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Influenza · COMBAT | 21 | 10 | 1.000 | 0.0050 | 1.000 | 0.0050 | 0.982-1.000 | +0.000 | 0.0010 | — | **not estimable** |
| Sepsis vs COVID-19 · COMBAT | 111 | 11 | 0.996-0.998 | 0.0050 | 0.982-0.985 | 0.0050 | 0.894-0.962 | -0.001 to +0.001 | 0.0010 | 0.0569-0.6144 | **underpowered** |
| SLE · GSE174188 (CD4) | 261 | 99 | 0.872-0.889 | 0.0050 | 0.762-0.774 | 0.0050 | 0.978-0.985 | +0.087 to +0.101 | 0.0010 | 0.0010 | estimable |
| COVID-19 · COMBAT | 110 | 10 | 0.862-0.884 | 0.0050-0.0100 | 0.500-0.504 | 0.7413-0.8657 | 1.000 | +0.116 to +0.138 | 0.0010 | 0.0010 | **underpowered** |
| COVID-19 · Stephenson | 115 | 29 | 0.845-0.882 | 0.0050 | 0.501-0.594 | 0.0149-0.2338 | 0.972-0.996 | +0.104 to +0.141 | 0.0010 | 0.0010 | estimable |
| COVID-19 · Ren | 182 | 25 | 0.812-0.854 | 0.0050 | 0.729-0.768 | 0.0050 | 0.964-0.982 | +0.100 to +0.164 | 0.0010 | 0.0010 | estimable |
| SLE · GSE174188 (all cells) | 261 | 99 | 0.819-0.832 | 0.0050 | n/a | n/a | 0.992-0.995 | +0.158 to +0.172 | 0.0010 | 0.0010† | estimable |
| COVID-19 · Ren (10x 5’ v2) | 161 | 25 | 0.712-0.729 | 0.0050-0.0149 | n/a | n/a | 0.954-0.977 | +0.227 to +0.262 | 0.0010 | 0.0010† | estimable |
| CMV · HIHA | 108 | 45 | 0.404-0.518 | 0.3831-0.8806 | 0.365-0.489 | 0.5274-0.9453 | 0.894-0.933 | +0.306 to +0.354 | 0.0010 | 0.0010 | estimable |

**Table 1. The nine comparisons, ordered by how well the recorded metadata predicts the phenotype label.** Generated from the screen output by `tools/build_table1.py`. Every comparison was run at five split seeds; each value is the range over those five, and a single value means all five agreed to the printed precision. "Minority" is the smaller of the two label groups. "Metadata AUC" uses all three variable groups together; "Collection AUC" uses the collection block alone, which is study design in the strict sense, and n/a means the comparison records no collection variable that varies. Every AUC is cross-fitted under the prespecified fixed-hyperparameter pipeline, which is the statistic each p-value beside it was computed against and the one both permutation tests are built from; the tuned nested-CV AUCs, the two column counts, V_D and p(V_D) are in Table S4, per seed. "+ over metadata" is the change in AUC when expression is added to the recorded metadata in one joint model, out of fold (Section 4.5, Table S5). Each p is one-sided against a permutation null built from that comparison's own metadata matrix. The two floors, 0.0050 and 0.0010, are 1/201 and 1/1001: the metadata-only AUC and V_D nulls use 200 permutations, the complete-pipeline tests 1,000. A dash under "p strat." means the collection-stratified permutation is uninformative because every stratum is label-pure. A dagger marks the two comparisons that record no collection variable at all, where the stratified permutation falls back to tertiles of log cells per donor: for those two it is a sample-quality-stratified permutation, and they are the same two rows showing n/a in the collection columns. Verdicts are read from the screen's own gates: **not estimable** when no collection stratum holds both labels, which is the single criterion, and **underpowered** when the minority group has fewer than 15 donors.

## 4.2. Metadata predictability in the CMV comparison, a negative control specified in advance

We chose the CMV cohort before computing anything, on its recorded structure alone. Its 108 donors were assayed on one chemistry and one suspension type, and the CMV status of a donor is a serological result rather than a recruitment criterion, so we expected the collection labels to be close to independent of who is positive. If these checks measure what they claim to measure, this cohort has to come out at the bottom.

The structure needs stating precisely, because the phrase "single-centre cohort" invites the wrong picture. This cohort is **not** one batch: the released metadata records 36 batch identifiers and 46 pool identifiers, and their crossing gives 46 strata, 17 of them holding a single donor. Neither is it donor-paired in any sense this analysis uses - each donor contributes one row with one label, and no within-donor contrast is formed anywhere in the paper. What makes it a negative control is not a simple design but a weak association between the collection labels and the phenotype label, which is exactly the quantity being screened.

It comes out at the bottom. Across five split seeds the metadata-only AUC for the full recorded set is 0.404 to 0.518, straddling chance, with permutation p 0.383 to 0.881 - significant in no seed. Collection variables alone give 0.365 to 0.489, p 0.527 to 0.945, and sample quality 0.453 to 0.507, p 0.478 to 0.716. The same 108 donors support an expression classifier at AUC 0.894 to 0.933, with both permutation tests at their 1/1001 resolution floor in every seed. That is the prediction, and it holds.

One exception within the comparison. Demographic variables alone reach a metadata-only AUC of 0.591 to 0.639, with permutation p 0.015 to 0.085; applying Benjamini-Hochberg [24] within each seed to that seed's 34 variable-group tests leaves $q = 0.020$ in two of the five seeds and 0.074 to 0.115 in the other three. So the demographic block is significant after correction in a minority of seeds, and we report it that way rather than as a uniform null. Sex and ethnicity are established correlates of cytomegalovirus seroprevalence in their own right, so an association there is as easily aetiology as recruitment. This is what the three-way split in Section 4.1 is for: it is a reason to read that block as case mix, and a reason not to let it stand for study design. It is also the reason the negative control is stated at the level of the collection block, where nothing is detected in any seed.

Two further points of interpretation. First, this is the concrete case for building each statistic's null from the same matrix rather than against a fixed reference. Read without its own null, the in-sample V_D of 0.468 looks like a strong association - 89 columns fit 108 donors closely whatever the labels are. Against the matched in-sample null it is 0.468 against a null mean of 0.496, with p(V_D, in-sample) 0.284 to 0.353. The cross-fitted pair, which is what p(V_D) means everywhere else in this paper, sits at its ceiling of 1.000 with $p = 1.000$.

Second, no detected association is not the same as no association. The metadata-only test has whatever power 108 donors and this matrix give it, and the matrix is wide: the collection block declares 80 one-hot columns but its centred rank is 45, and the full recorded block declares 89 columns at rank 54. Column count is not effective dimension, and neither number should be read as the number of independent directions the phenotype was tested against. What the comparison supports is that the recorded collection labels carry no detectable information about CMV status at this sample size, which is what a negative control is for.

### 4.3. Variation in metadata predictability across cohorts

Ordered by the metadata-only AUC, the nine comparisons run from 0.404 to 1.000 (Figure 3A, Table 1), with no gap that would justify a threshold. Positions are stable across seeds: excluding the CMV control, whose near-chance value moves 0.114 between seeds precisely because it is near chance, the widest five-seed spread is 0.042. Eight of the nine have a significant association with the recorded metadata in every seed, and the count is still eight after Benjamini-Hochberg correction within each seed, whether the family is that seed's nine all-metadata tests or all 34 of its variable-group tests. The CMV negative control is the ninth.

Two comparisons show why the metadata-only AUC rather than V_D carries the test (Section 2.2). In COVID-19 Ren (26 metadata columns, 182 donors) and COVID-19 COMBAT (7 columns, 110 donors) the cross-fitted V_D is unstable across seeds, running from 0.847 to its ceiling of 1.000 in Ren and 0.906 to 1.000 in COMBAT. When it reaches the ceiling the matched null reaches it too, so p(V_D) is 1.000; in the other seeds the same comparison gives p(V_D) = 0.005. The metadata-only AUC on the very same matrices is stable by contrast, 0.812-0.854 and 0.862-0.884, significant in every seed (p = 0.005 and 0.005-0.010). Counting by p(V_D) alone would give six of nine rather than eight; we report both counts and use the better-powered one.

Under the primary, collection-only definition the count is smaller, and we give it rather than leave the union to stand for study design. Seven of the nine comparisons record any collection variable at all; in four of those seven the collection-only AUC is significant in all five seeds, and two of those four are the comparisons flagged as not estimable or underpowered. Among comparisons that are both estimable and record collection variables, two of four are significant: lupus in GSE174188 CD4

(0.762-0.774) and COVID-19 in Ren (0.729-0.768). Two comparisons record no collection variable that varies - lupus in GSE174188 all cells, and the single-assay Ren subset - so their stratified permutation conditions on tertiles of log cells per donor instead. For those two the test is sample-quality-stratified, and Table 1 marks them rather than reporting them as collection-stratified. This difference is the point of separating the groups: much of what the union detects is carried by sample quality and case mix, which are not experimental design and which carry their own interpretations (Section 4.1).

Grouping the metadata variables shows that similar overall strength can come from different parts of the collection process (Figure 3B). The clearest case is a pair of COVID-19 cohorts of similar size and closely comparable metadata-only AUC (Figure 3C):

- **Stephenson** (n = 115, recorded-metadata AUC 0.845-0.882). Sample-quality variables alone reach 0.865-0.884 (p = 0.005 in every seed), while collection variables reach only 0.501-0.594 and are not significant in every seed (p = 0.015-0.234). The metadata-phenotype association is carried mainly by sample quality.
- **Ren** (n = 182, recorded-metadata AUC 0.812-0.854). The mirror image: collection variables reach 0.729-0.768 (p = 0.005 in every seed), while sample quality reaches only 0.590-0.632 and is not significant (p = 0.050-0.154). The association is carried mainly by recruiting hospital and sub-study.

Similar overall metadata-only AUCs therefore concealed different patterns of association. The Stephenson result motivates closer examination of library-quality measures, and of whether harmonised preprocessing changes the estimate; the Ren result motivates examination of recruiting site and sub-study composition, where the relevant question is whether donors exist under overlapping collection conditions. A single number would call the two cohorts equally affected and point both at the same follow-up. This is also the direct answer to the point that V_D depends on how the metadata matrix is coded: it does, and the response is to report the grouping rather than to hunt for a coding-free number.

The Ren cohort also shows how an incomplete set of metadata variables misleads. With only sequencing chemistry recorded as collection metadata, the collection-only AUC is 0.553 - apparently untroubled by site. Recovering the recorded hospital (12 centres, five of them case-only) and sub-study identifier (six source studies) raised that group to 0.750 and the full set from 0.758 to 0.818. The problem was there and invisible. These checks are only as good as the metadata they are given, and a reassuring result computed on thin metadata is weak evidence.

### 4.4. Free and collection-stratified permutation results

**Where the stratified test is uninformative.** The COMBAT influenza comparison has 21 donors and metadata that separates the labels exactly (metadata-only AUC 1.000, V_D 0.000). Every collection stratum is label-pure (Figure S4), so within-stratum permutation leaves all labels unchanged and the reference distribution is degenerate; we report the stratified test as uninformative, with p = NA. The free permutation is still defined and rejects at the 1/1001 floor in all five seeds, behind an expression AUC of 0.982 to 1.000. Reported alone, that reads as a comfortably validated classifier. What the two results together say is that recorded metadata and the phenotype cannot be separated in this comparison with the recorded data, and that the test which would have detected it is the one that has no reference distribution here. We report the comparison outside the main ordering.

**Where the two tests point in different directions.** In the COMBAT sepsis-versus-COVID-19 comparison (n = 111, minority class 11), metadata predicts the phenotype almost perfectly, at 0.996 to 0.998 across seeds. The free permutation yields p = 0.001 in every seed; the collection-stratified permutation yields p = 0.057 to 0.614, significant in none (Figure 3D). The classifier therefore shows evidence of an overall association between expression and the phenotype, but no evidence of discrimination beyond what the specified collection strata retain. The two results are not contradictory - the tests state different nulls - but a report carrying only the first would read as a validated classifier, and that reading is not supported here.

Two cautions on this row. The smaller class holds 11 donors, so the stratified test has little power and its failure to reject is not a demonstration that the discrimination is entirely collection-driven. And the fixed-hyperparameter pipeline scores 0.011 to 0.066 AUC above the tuned one here, the largest gap among the eight comparisons that return a number. That is not a feasibility failure - the fixed pipeline is the inferential statistic, and it is what both permutation tests are built from -

but with 11 donors in the minority class a reader should be able to see how much of the headline AUC depends on model selection.

### 4.5. What expression adds over the recorded metadata

Reporting the metadata-only AUC beside the expression AUC says whether each predicts the phenotype. It does not say whether expression carries information the recorded metadata does not already have, which is the question a reader of a patient-level classifier has. We fitted the third model: inside each training fold, expression is reduced to principal components on the training donors, the metadata matrix is built from the same donors, and one classifier is fitted on the concatenation. All three arms run the identical fixed-hyperparameter specification through identical folds, so a difference between them is a difference of information (Table S5).

In seven of the nine comparisons the increment over metadata alone is large: adding expression raises the AUC by 0.087 to 0.354 across five seeds, and in each of those the joint model is within 0.023 of the expression-only model, so the metadata contributes little once expression is present. Lupus in GSE174188 CD4 gains 0.087 to 0.101 over metadata alone; the single-assay Ren subset, which records the least collection metadata, gains the most at 0.227 to 0.262.

The two exceptions are the two comparisons already flagged. In sepsis-versus-COVID-19, metadata alone reaches 0.996 to 0.998 and adding expression changes the AUC by $-0.001$ to $+0.001$; expression adds nothing measurable to what the recorded metadata already provides, while metadata adds 0.036 to 0.104 over expression alone. In influenza both models are at 1.000 and the increment is exactly zero in every seed. The increment therefore agrees with the permutation results, and agrees with them for a reason that requires no permutation: these are the two comparisons where the recorded metadata already accounts for the separation.

CMV runs the other way and is worth stating. Its metadata-only AUC is at chance, its expression AUC is 0.894 to 0.933, and the joint model is *lower* than expression alone, by 0.069 to 0.136. Appending 89 uninformative one-hot columns to 50 principal components costs discrimination rather than adding it - the same property that makes residualisation on that matrix expensive (Sections 3.5, 5.3).

The increment is reported as a descriptive quantity over five seeds, with no p-value attached. A significance test for it would have to hold the metadata association fixed while permuting, which is the collection-stratified permutation already reported in Table 1 on the statistic it is already reported for.

### 4.6. Scope of the ordering

The ordering says how well recorded metadata predicts the phenotype label. It does not show that any particular published classifier depends on that metadata, because it does not test whether the classifier uses the metadata-related part of X - the D $\rightarrow$ X $\rightarrow$ $\hat{\mathrm{Y}}$ path in Figure 1. Sections 5.2 to 5.4 test that path directly in the two lupus cohorts, and reach a more equivocal answer than the ordering alone would suggest.

## 5. Whether the classifier uses the metadata-related structure: two lupus cohorts

The ordering measures the D-Y arrow. This section follows the D $\rightarrow$ X $\rightarrow$ $\hat{\mathrm{Y}}$ path in the two lupus cohorts for which complete donor metadata is held, using frozen Geneformer [4] embeddings and two expression pseudobulk representations under one fold-contained pipeline. GSE285773 [3] serves as the independent transfer source. Figure 4A shows how cases and controls are distributed across the recorded design in both cohorts; Figure 4B shows how well the recorded metadata alone predicts disease status.

### 5.1. All three representations encode batch

In GSE174188 the three representations classified disease at AUC 0.982, 0.977 and 0.975. The same representations identified processing wave, one against the rest, at 0.9997, 0.999 and 0.997. In GSE135779 disease AUCs were 0.870, 0.933 and 0.945, while the best single-batch AUCs were 0.993, 0.961 and 0.961 (Figure 5A).

This shows that the representations contain batch information. It shows nothing more. A representation that encodes batch can still carry disease information, and a batch-aware representation need not produce a batch-driven decision rule. Batch predictability is therefore not a validity check on its own, and we report it below only with that scope attached.

### 5.2. Batch predictability did not imply sensitivity to batch adjustment

GSE135779 provides the counterexample directly. In that cohort, Geneformer embeddings predicted the most distinguishable batch at AUC 0.993 and both pseudobulk representations at 0.961, close to perfect batch recognition, whereas batch alone predicted disease status at AUC 0.499. Batch residualisation then changed disease AUC by +0.002, −0.002 and −0.009 (Figures 5B, 6C) - three shifts well inside the split-to-split spread, so the classifier's dependence on batch is small rather than demonstrably zero. Strong batch information in a representation therefore did not imply a substantial change in disease discrimination under the batch adjustment evaluated here, and "the representation encodes batch" does not license "the classifier uses batch" on these data.

### 5.3. Residualisation and restriction answer different questions

Restriction asks whether the separation survives among donors collected under overlapping conditions. Residualisation deletes the part of the data that a chosen set of metadata variables can predict. When metadata and phenotype are collinear these are not the same question, and we compare them to show they are not interchangeable, not to rank them.

In GSE174188, ridge residualisation on processing-wave fractions reduced the three representations from 0.982, 0.977 and 0.975 to 0.714, 0.747 and 0.735. Restriction to the one large near-balanced wave, compared against size- and label-matched random donor subsets - the comparison that separates the cost of conditioning from the cost of using fewer donors - cost only 0.014 to 0.049 (Figure 6B). The paired difference between the two losses was 0.186 to 0.255 across six comparisons, with every descriptive interval above zero.

In GSE135779, removing the single all-case batch left 36 donors with both labels in every remaining batch and changed matched performance by at most 0.012. Five ways of removing the recorded variables give five different answers for the same cohort (Figure 6A). Projecting out the complete set with ridge collapsed all three representations to 0.516, 0.522 and 0.520. A random-forest residualiser removed less (down to 0.748, 0.787 and 0.795), and overlap weighting and batch location adjustment less still (retaining 0.859 to 0.937). Every full-set removal took away far more than the batch-only control.

The algebra explains this without needing a technical story. When design predicts the diagnosis, the part of the data that design can predict necessarily includes diagnosis-related variation. Removing it removes disease signal regardless of where that signal came from. A large loss after residualisation is therefore **not** evidence that the original AUC was spurious. It is equally consistent with removing a shortcut and with removing real biology, and the size of the loss does not distinguish them.

The CMV comparison settles part of what residualisation loss actually measures. There, no association is detectable between the full recorded metadata and the phenotype: metadata-only AUC 0.512 under this pipeline, permutation p 0.383 to 0.881 across five seeds, and $p(V\_D) = 1.000$. Residualising its representations on those same variables still reduces classification AUC from 0.925 to 0.599, a loss of 0.326 (Figure 6C). The discrepancy is worth stating plainly: an adjustment made for a metadata-phenotype association that no test detects removes a third of the separation anyway.

We tested the obvious explanation rather than asserting it. The matrix declares 89 one-hot columns for 108 donors, at a centred rank of 54, and the collection block alone declares 80 columns at rank 45; a projection of that reach takes out most of any sample. The simulation in Section 3.5 generates a metadata matrix independent of the phenotype by construction and residualises with the identical procedure. A balanced matrix of the same width (92 columns) removes 0.159, and the loss is not monotone in width, peaking at 0.333 for 48 columns. A matrix built to the ragged shape of the real batch-by-pool crossing - many levels holding one or two donors - removes **0.363 on average, central 95% 0.272 to 0.464**, which contains the 0.326 observed here (Figure S2).

So the observed CMV loss is the size that this adjustment procedure produces on a matrix of this shape when there is provably nothing to remove. The conclusion is about residualisation, not about the cohort: the size of a residualisation loss depends on properties of the adjustment matrix that have nothing to do with confounding, and cannot be read as a measure of technical contamination.

It is tempting to read the pair of values from residualisation and restriction as a range bounding the true biological effect. It is not one. There is no proof that the two estimates bracket any underlying quantity. They are two estimates made under two different sets of assumptions, and their disagreement is informative precisely because neither is identified.

### 5.4. Cell-type composition reproduces the same pattern

Aggregating the same GSE174188 CD4 cells into naive, effector-memory, regulatory and unassigned fractions gave disease AUC 0.899 to 0.915. The same four-number summaries predicted processing wave at 0.803 to 0.872. Wave residualisation reduced them to 0.689 to 0.709 (Figure S5A), and within-wave restriction against matched subsets cost 0.027 to 0.062, with every interval crossing zero (Figure S5B). A four-dimensional biological summary therefore reproduces the whole pattern. That places the problem in the cohort rather than in foundation-model embeddings, and is consistent with composition being a strong patient stratification baseline in its own right [26].

### 5.5. Source restriction and cross-cohort discrimination

Restricting the training set to one processing wave is the obvious remedy: train where the design is uniform, then test elsewhere. It does not work. Training on all 261 GSE174188 donors and testing in the independent 26-donor GSE285773 cohort gave 0.900, 0.944 and 0.969. Restricting the source to the dominant wave lowered every one of them, and did no better than a random subset of the same size (Figure 7). Within GSE174188, training on waves 2 and 3 and testing on wave 4 gave 0.842, 0.811 and 0.806, and the reverse direction 0.919, 0.926 and 0.821. More varied source donors transferred better than the apparently cleaner restricted source.

Transfer in the other direction, from the 26-donor cohort to the 261-donor target under strict source-only rules, gave 0.884 [0.843-0.920], 0.919 [0.884-0.946] and 0.926 [0.895-0.953]. Both pseudobulk representations beat Geneformer after paired DeLong testing [23] and Benjamini-Hochberg adjustment [24] ($q = 0.0151$, $q = 0.000603$).

Whether a more expressive donor-level aggregator changes this is a separate question from the one this paper asks, and we report it in the supplement rather than the main text. Briefly: three learned pooling methods on the same frozen embeddings never beat ordinary mean pooling, and every significant paired difference was negative (Figure S6, Table S6). Adding capacity fitted the source without producing a representation that travels.

## 6. Integrating metadata assessment and predictive validation

These are not a validity standard. We have not shown that they are sufficient, that they are minimal, or that passing them licenses a causal claim. They are five checks that look at different arrows in Figure 1 and that fail in different ways. Their value is that they can disagree with each other.

| # | Check | Shows | Does **not** show |
|---|---|---|---|
| 1 | Metadata-only AUC by variable group, cross-fitted V_D, p(V_D) against the cohort's own null, overlap counts | Whether recorded metadata predicts the phenotype label, and which variable groups do | That the classifier uses that metadata; anything about unrecorded structure |

| # | Check | Shows | Does **not** show |
|---|---|---|---|
| 2 | Complete-pipeline permutation, both free **and** collection-stratified | Free: evidence of an expression-phenotype association of any origin. Collection-stratified: evidence of discrimination beyond what the collection strata retain | Neither identifies a mechanism, and neither certifies the absence of leakage — a label-dependent step taken before the permutation is present on both sides. The stratified test conditions on collection variables only, so it does not cover sample-quality or demographic association inside a stratum (Section 3.4), and it is uninformative when every stratum is label-pure, where the free test is still defined |
| 3 | How well the representation predicts batch | That the representation encodes collection information | That the phenotype decision rule depends on it — Section 5.2 is a counterexample |

| # | Check | Shows | Does **not** show |
|---|---|---|---|
| 4 | Residualisation **paired with** restriction and matched donor controls | How far two non-equivalent adjustments disagree | Which adjustment is right; bounds on a biological effect |
| 5 | Source-only external target | Whether patient ranking survives a change of collection process | Calibration or clinical usefulness; and it carries much less weight when the target was collected the same way, since a shared bias reproduces rather than cancels (Section 3.3) |

Checks 1 and 2 cost minutes of CPU on a donor-level table and are the ones we would run first on any new cohort. Check 5 is the only one that speaks to transport, and no combination of the internal checks substitutes for it.

**What a clinician should take from this.** An internal cross-validated AUC measures discrimination under the conditions the cohort was sampled in, which is a real quantity and the right starting point. Two further things extend it. A metadata-only model says whether the recorded collection differences alone could produce that discrimination, and a conditioned validation - the collection-stratified permutation, or restriction to overlapping strata - says whether it exceeds them. An independent-cohort evaluation then addresses transfer to a new setting, and carries much less weight when the second cohort was assembled the same way as the first, because a shared bias reproduces rather than cancels. A classifier reported with all three is worth taking seriously as a candidate diagnostic; one reported with the internal AUC alone has not yet been separated from its cohort.

Applied to the nine comparisons, the checks give three groups rather than a pass and a fail. In the first, no association with the recorded metadata is detectable and internal discrimination is not undermined by anything visible here - which is weaker than saying it is biological, since unrecorded structure remains possible; only CMV is here. In the second, the metadata predicts the phenotype strongly and yet the collection-stratified permutation is still rejected, so the discrimination exceeds what the collection strata retain; both lupus comparisons and all three COVID-19 cohorts are here. In the third, the stratified permutation is either not rejected or uninformative, and no internal analysis settles the question; sepsis-versus-COVID-19 and influenza are here. Only the third group is a stop condition, and it is a statement about the cohort, not about the method being evaluated.

### 6.1. Three comparisons traced through the tree

Figure 8 traces four comparisons through the decision tree with the evidence for each branch stated. Three are given in full here; the fourth, sepsis versus COVID-19, is described in Section 4.4. The strata, the metadata matrix and the classifier are all imported from the screen rather than rebuilt for the walkthrough, so Q2 and Q4 ask about the same partition, the same columns and the same fitted pipeline.

Every branch is written to output what can be estimated and what is still missing, not a recommended remedy. That distinction is what the previous version of this tree got wrong, and Section 8 records the correction: a branch may say that no collection association was detected, and it may say that a conditioned estimate is unavailable, but it cannot on that basis instruct anyone to residualise. Whether to adjust is decided by the target quantity and the scientific question, and the evidence here bears on it without settling it.

**COMBAT influenza (n = 21) stops at Q0.** Its stratum variable is the recording institute, which gives two strata, *neither of which contains both a case and a control* (Figure S4). Within-stratum permutation therefore leaves every label unchanged and the stratified test is uninformative. That, and only that, is the stopping condition: the comparison offers no donors under overlapping collection conditions, so no conditioned estimate can be formed from these data and no route in the tree is available. Its metadata-only AUC is also 1.000, which is consistent with the same structure but is not what triggers the stop - a finite-sample AUC of 1.000 records that the fitted model separates these donors completely, and is not on its own a proof of non-identifiability. The fixed and tuned pipelines agree closely here, 0.982 to 1.000 against 0.964 to 1.000 across seeds, so nothing about the stop depends on which is read. Note what the free permutation would have said alone: p = 0.001 in every seed, which reads as a validated classifier and is the wrong description of this comparison.

**CMV HIHA (n = 108) reaches Route A.** The stratified permutation is defined - 46 strata from batch and pool, 21 of them holding both labels - and both permutation tests reject at the 1/1001 floor. At Q3 no association with the recorded metadata is detected: metadata-only AUC 0.512 under the walkthrough's pipeline and 0.404 to 0.518 in the main screen, not significant against its own permutation null in any seed. Route A therefore outputs *no detected collection association; the unadjusted estimate stands as the reported quantity.* It does not output a recommendation to residualise, and this comparison is the reason. Residualising here costs 0.326 AUC (Section 5.3) for an association no test detects, and the Section 3.5 simulation shows a matrix of this shape costing 0.363 with the association absent by construction. A block that is not detected is not thereby shown to be negligible - the test has whatever power 108 donors afford - but neither is undetectability a reason to adjust. Whether to adjust is a decision about the target quantity, and Route A reports the evidence rather than making it.

**COVID-19 Ren (n = 182) reaches Route C.** Both permutation tests are rejected and the association with the recorded metadata is strong: metadata-only AUC 0.832 under the walkthrough's fixed-penalty pipeline and 0.812 to 0.854 under the main one. Label-mixed strata exist, so Q4 is reached. Q4 asks whether restriction is *feasible*, and this is the question the earlier version of the tree left open: it asked only whether a mixed stratum existed, which let the same comparison be assigned to a restricted estimate at Q4 and then described as needing redesign a paragraph later. The requirement is now fixed in advance - a comparison takes the restricted route only if some stratum holds at least five donors of each class *and* at least 40 donors in total, the same power floor used everywhere else in this paper - and each comparison therefore has one output.

Ren does not meet it. Exactly one stratum is large enough to cross-validate at all, and it holds 18 donors; the restricted estimate there is AUC 0.992 against 0.733 in size- and composition-matched random subsets, but the matched-subset standard deviation is 0.145, which is the width of the whole effect being estimated. Residualising the 26-column matrix costs 0.083, from 0.967 down to 0.884. The tree therefore outputs Route C: the association is present, both adjustments are available, they disagree in direction, and neither is estimable with useful precision on these donors. Both numbers are reported, no bound is claimed, and the outstanding requirement is donors collected under overlapping conditions rather than a further analysis of these ones.

**Restriction is rarely available at these strata, and the tree now says so.** CMV has 21 label-mixed strata and *none* with five donors of each class. Ren has one usable stratum of 18 donors. Route B is drawn as though restriction were a practical alternative to residualisation; on real cohorts with realistically fine strata it frequently is not. We state that rather than coarsening the strata until Route B becomes available, which would make the tree unfalsifiable.

# Discussion

## 7. Implications for patient-level single-cell evaluation

Across the nine comparisons, phenotype predictability from recorded metadata was measurable, graded and traceable to specific variable groups. It ran from a metadata-only AUC of 0.404 to 1.000, more than half the usable scale, and each cohort's position was stable across split seeds. The *raw* residual label variance, by contrast, is not comparable across cohorts without its own null: an in-sample $R^2$ ranks the least affected comparison in this set as the third most affected, purely because its metadata matrix is wide relative to its donor count.

Similar overall metadata-only AUCs did not imply similar underlying associations. In the Stephenson COVID-19 cohort, sample-quality variables reached AUC 0.865 to 0.884 against 0.501 to 0.594 for collection variables; in Ren, collection variables reached 0.729 to 0.768 against 0.590 to 0.632 for sample quality. The two overall figures differ by about 0.02. The Stephenson result motivates closer examination of library-quality measures and of whether harmonised preprocessing changes the estimate; the Ren result motivates examination of recruiting site and sub-study composition, where restriction to balanced strata is the relevant question and, as Section 6.1 shows, is not available at these donor counts. Reporting one number hides the part of the answer a reader can act on.

The two permutation tests have the widest reach beyond this dataset. Free label permutation is standard practice and answers what it is asked: whether expression and the phenotype are associated at all. It cannot answer whether a classifier exploits the collection process, because the shuffle destroys that process along with everything else. Stratified permutation is not new - Chaibub Neto and colleagues introduced it for exactly this purpose [30] - and what these cohorts add is what the two tests return when the acquisition structure varies. In sepsis-versus-COVID-19 the free test rejects in every seed and the stratified test in none; in influenza the stratified test has no reference distribution at all; in the other seven they agree. A study using label permutation to argue that a patient-level classifier is valid should report the stratified version beside it, or state that no label-mixed stratum exists and that the test is uninformative. That statement is itself the more informative outcome.

The stratified test has a limit of its own, and it is better published as a number than as a caveat. It conditions on collection strata, not on the whole recorded matrix, so a sample-quality variable that stays associated with the phenotype inside a stratum can carry a classifier past it with no disease effect present; Section 3.4 measures how often. Rejecting it therefore rules out one specific explanation - these strata - and not the general one. Detecting a conditional association is not a certificate of disease biology.

The simulation adds a caution about external validation that we did not anticipate. When the external cohort shares the source's collection mechanism, external AUC rises with confounding rather than falling. An external result is evidence about transport only in proportion to how independent the second collection process really is, and being a different accession number does not establish independence.

For representation research the implication is narrower than a leaderboard result but harder to work around. In our lupus data pseudobulk beats frozen embeddings in strict source-only transfer, and added pooling capacity does not change that. The more durable point is that internal accuracy in these cohorts is not a measurement of the property that matters, and no amount of model development fixes a cohort in which design predicts the diagnosis. That is a recruitment and data-release problem. The fix is to record and distribute collection metadata, and to build cohorts in which cases and controls appear together within batches - not to build a better encoder.

None of these results supports a diagnostic claim. The external AUCs reported here rank 261 or 26 donors and are not calibrated probabilities. What the analysis supports is a reporting discipline. A patient-level classifier should be published with three things. First, the metadata-only AUC of its development cohort **by variable group**, so a reader can see whether the association runs through collection, case mix or sample quality. Second, both permutation tests, or a plain statement that the stratified one is uninformative because every stratum is label-pure. Third, an external result on a cohort collected through a different route, fitted on the source cohort alone. None of that requires a new method. It requires releasing the metadata and running two cheap checks.

## 8. Limitations

These analyses address association with recorded metadata rather than causal biological specificity. Unmeasured collection structure - referral patterns, institutional treatment protocols, sample handling not captured in released metadata - is invisible to every quantity reported here, so a low metadata-only AUC is weak evidence of an unaffected cohort rather than strong evidence of one. What that costs is visible within the Ren cohort itself: a metadata set that omits the recruiting hospital gives a collection-only AUC of 0.553 where the completed set gives 0.750, on the same donors.

**The collection-stratified permutation conditions on a subset of the recorded metadata.** Its strata come from collection variables; sample-quality and demographic columns are in the metadata matrix but not in the strata. Section 3.4 measures the consequence directly: with a sample-quality variable associated with the phenotype inside a stratum and driving expression, the test rejects at rates rising to 95% with no disease effect anywhere in the generating model. Rejecting it is therefore evidence of discrimination beyond the collection strata, not evidence of a disease-specific signal, and not a demonstration that some part of the discrimination is free of association with the recorded metadata. Building a null that conditions on the full matrix, including its continuous columns, would require a conditional randomisation scheme we do not develop here. That is the single largest methodological gap in this paper.

**The three variable groups are not equivalent, and only one is study design in the strict sense.** Demographics can be case mix or genuine risk; sample-quality summaries sit downstream of the assay but can also sit downstream of the disease. We therefore report the collection block separately everywhere and treat it as primary for any claim worded as being about study design. The headline count differs between the two definitions, and both are given.

V_D is linear by construction and depends on how the metadata matrix is coded. Reporting nonlinear metadata-only classifiers, the variable grouping and the matrix's own null reduces but does not remove this. The nonlinear arms use two tree ensembles and are not an exhaustive comparison of learners. Column count is not effective dimension either: the CMV collection block declares 80 one-hot columns at a centred rank of 45, and no reading of these results should treat a column count as a number of independent directions.

**Residualisation results are conditional on one adjustment procedure.** Section 3.5 generates a metadata matrix independent of the phenotype and still measures a loss of 0.363 at the CMV matrix's shape, which contains the 0.326 observed. That establishes that a loss of this size carries no information about contamination; it does not establish what a different residualiser would do. Ridge at a fixed penalty on a training-fold standardised matrix is the only adjustment characterised at this depth, and the random-forest, overlap-weighting and location-adjustment arms in Section 5.3 give materially different answers on the same cohort.

**The decision tree organises the evidence; it is not a validated instrument.** Two of its properties are worth stating because they are easy to assume otherwise. A null result at Q3 reports that no collection association was detected and leaves the unadjusted estimate standing; it does not recommend an adjustment, and CMV is the case that shows why one should not be inferred from a null test. And Q4 carries an explicit feasibility requirement, so a comparison with a label-mixed stratum too small to support an estimate takes Route C rather than receiving one. Neither the branch structure nor the thresholds have been validated against an external criterion, and no result in this paper depends on the tree.

All nine comparisons were run at five seeds, and every value we quote is the range over those five. Three comparisons have fewer than 15 donors in the minority class. We flag rather than exclude them, but no conclusion should rest on them.

The nine comparisons come from five datasets and are **not independent**. Comparisons drawn from the same dataset share donors, so the count of eight in nine is a count of comparisons, not of independent replications.

Matched random subsets separate the cost of restriction from the cost of using fewer donors without identifying a causal technical effect, so the restriction results show non-attributability rather than that retained signal is biological. Coverage is four diseases from five cohorts, all peripheral blood, all public, all curated under one schema; tissue-resident and non-blood settings are untested, so the cross-disease claim here is not a cross-tissue one. Geneformer pretraining overlap with individual benchmark cells was not tested.

Finally, these are checks. They order cohorts and point at variable groups. They do not identify a causal effect, and no claim here depends on their doing so.

# Methods

## 9. Materials and methods

### 9.1. Cohorts and the unit of analysis

Seven public datasets were used, in two groups. Table S7 is the registry. It gives, for each dataset, the identifier, the donor count reported by the source, and the donor, case and control counts actually modelled - these differ, because donors below the minimum cell count or carrying a label outside the contrast are dropped. Five entries were verified against the CELLxGENE Discover API on 2026-09-07; the two used only in Section 5 come from GEO and are marked as not API-verified.

**Five datasets enter the main screen**, all obtained through the CELLxGENE Discover curation API: the CD4-positive alpha-beta T-cell subset of GSE174188 (261 donors; 162 cases, 99 controls) [2], the Ren COVID-19 atlas (196 donors) [39], the Stephenson COVID-19 cohort (120 donors) [40], the COMBAT consortium (124 donors) [41] and the HIHA cytomegalovirus cohort (108 donors) [42]. Nine case-control comparisons are formed from these five, so the comparisons are not independent of one another; Table 1 gives the donor overlap.

**Two further lupus datasets are used only in the downstream analyses of Section 5**, not in the screen: GSE135779 (44 childhood donors; 33 cases, 11 controls) [1] and GSE285773

CD4-positive T cells (26 donors; 16 cases, 10 controls) [3], the latter as the independent transfer target.

Registry entries, dataset identifiers, donor counts and download sizes were verified against the API on 2026-09-07 and are released in `cohorts/registry.yaml`.

The donor was the independent unit in every supervised analysis. Cells from one donor never crossed a training/test partition.

### 9.2. Donor-level tables and the metadata variables

Each cohort was streamed from its h5ad in 100,000-cell chunks and reduced to one row per donor. Expression was aggregated to log1p CPM pseudobulk. The layer used for aggregation was chosen by sniffing for integer counts rather than assuming a layer name.

Design variables were drawn from schema-guaranteed observation fields (`donor_id`, `disease`, `sex`, `development_stage`, `assay`, `tissue`) plus cohort-specific collection variables detected by pattern from the distributed object. Detection patterns cover batch, pool, run, lane, chip, site, centre, city, region, province, hospital, clinic, study, sub-study, source, dataset and donor source. Ages given as strings ("25-year-old stage", "fifth decade stage", "90 year-old and over stage") were parsed to years.

Covariates that are consequences of the disease were excluded by an explicit blocklist matching severity, outcome, comorbidity, medication, sample timing, symptom, diagnosis, stage, grade and WHO score. Supplementary Table S2 lists, for every cohort, which columns entered the metadata matrix, in which group, and which were excluded and why.

Variables were assigned to three groups. **Sample quality**: cells per donor, mean UMI per cell, mean genes per cell, aggregate mitochondrial percentage, and their logs. **Demographic**: age in years, sex, ethnicity. **Collection**: every detected `batch__*` variable, assay and suspension type.

### 9.3. Residual label variance

For centred phenotype-label vector $y$ and recorded metadata matrix $\mathbf{D}$ including an intercept, V_D = 1 − $R^2$(y ~ D). Two versions are reported. The in-sample version uses ordinary least squares on all donors and is reported for comparison. The cross-fitted version replaces the fitted values with out-of-fold predictions from five-fold `KFold` with shuffling, seeded by the run seed, and is the version used everywhere a number is interpreted. Negative $R^2$ values were clipped at zero before subtraction.

### 9.4. The metadata matrix's own permutation null

For each design matrix, the phenotype-label vector was permuted 200 times. Each permuted label was pushed through the **same five folds** used for the observed value, so the null is a distribution of the cross-fitted statistic, not of a different one. The reported p-value is one-sided, $(1 + \#\{\text{null} \leq \text{observed}\}) / (200 + 1)$, and answers: how often does a random phenotype label leave at most as much unexplained variation as the observed one? The null mean and its 2.5th percentile are reported alongside.

The metadata-only AUC is tested the same way, with 200 permutations of the phenotype label through the identical fixed-hyperparameter pipeline (`p_design_auc` in Table S4); that is the primary test, for the reason given in Section 2.2. The in-sample V_D is released with its own in-sample null, as `V_D_insample` and `p_V_D_insample` in Table S4. The two are separate quantities and are never mixed: p(V_D) always refers to the cross-fitted pair. This null is what makes either version readable, because its location depends on the number of metadata columns relative to donors.

### 9.5. The fitted pipelines

**Four analyses fit classifiers, and they are not one pipeline.** The multi-cohort screen, the decision-tree walkthrough, the calibration simulation and the GSE174188/GSE135779 deep dive each have their own settings, and Table S8 lists all of them. That table is generated by `tools/build_hyperparameter_table.py` from the `PipelineSpec` and `ForestSpec` objects in `scripts/cohorts/pipeline_core.py` that the runners themselves import, so no setting can be described here as something other than what ran. The paragraphs below state what differs; Table S8 is the authority on every value.

*Shared by the screen, the walkthrough and the calibration simulation.* One classifier specification, `SCREEN_FROZEN`, differing only in repeat budget: balanced logistic regression (liblinear, `max_iter`

5,000) at fixed inverse regularisation 1.0, stratified five-fold outer splits, the 4,000 highest-variance features selected on the training donors, and a PCA to 50 components (or fewer when donors are limiting) fitted on those donors and applied to the held-out ones. The screen runs one repeat, the walkthrough twenty. A tuned variant of the same specification, with the inverse regularisation selected from $10^{-4}$ to $10^{4}$ in decade steps by three-fold resampling inside each outer training set over five repeats, is reported beside the fixed one as a description; the fixed one is the inferential statistic, because it is what both permutation nulls refit.

*Metadata matrices.* The metadata matrix is small and dense, so no feature filter and no PCA are applied to it. Numeric variables are median-imputed from the training donors of each fold and standardised there; categorical variables are one-hot encoded from the level set observed in those training donors, dropping the first level, and a level seen for the first time in a held-out donor is encoded as the reference level. An earlier version of this screen built the whole matrix once on all donors before cross-fitting. No label is involved either way, and the change moves no headline count: the metadata-only AUC moves by at most 0.012 anywhere in the screen, and both the count of eight in nine and the count of four in seven are unchanged. It is corrected because the description in this section was otherwise not true of the code. The in-sample V_D is still defined on the whole-cohort matrix, as an in-sample statistic must be, and `n_design_feature` still reports that matrix's width.

*The lupus deep dive.* GSE174188 and GSE135779 predate the screen and run a wider budget on a single dataset: balanced logistic regression (liblinear, `max_iter` 20,000) with the same inverse-regularisation grid selected by five-fold resampling inside each outer training set, and 20 repeated stratified five-fold donor splits with integer seeds 20260801-20260820. Out-of-fold predictions are pooled within each repeat; we report the mean AUC and the 2.5th-97.5th percentile of repeat-level AUCs. That range is a descriptive split-sensitivity range. It is not a confidence interval, and it does not treat correlated folds or repeats as independent [31].

**Nonlinear metadata-only arms.** Two tree ensembles rather than a wider search, because the question is whether a *low-dimensional* metadata-phenotype relationship is being missed by a linear model, not which learner maximises AUC. The screen's forest uses 500 trees, unlimited depth, minimum leaf size 2, balanced class weights; the deep dive's uses 256 trees, maximum depth 4, minimum leaf size 3. Histogram gradient boosting, in the deep dive only, uses 150 iterations, learning rate 0.05, seven terminal leaves, minimum leaf size 5. All are fitted inside every outer training fold with fixed hyperparameters and are not selected against held-out donors: a tuned nonlinear arm would not be comparable with the tuned linear one under the same permutation null.

### 9.6. The two permutation tests

Both permutation tests and the observed statistic they are compared against run one identical fixed-hyperparameter pipeline, refitted from the raw donor-by-gene matrix on every permutation: within each training fold, the 4,000 highest-variance genes are selected on the training donors, a standardiser and a PCA to 50 components (or fewer when donors are limiting) are fitted on those donors and applied to the held-out ones, and a balanced logistic regression at fixed inverse regularisation 1.0 is fitted, with a single cross-fitting repeat. Every step that touches expression is therefore inside the fold, so the reported AUC is genuinely out-of-fold and the null refits the whole pipeline rather than reusing one representation. Freezing the regularisation is still necessary: comparing a tuned observed statistic against an untuned null biases p-values downward.

Every representation step is refitted inside the fold on every permutation because the code arranges it, not because the permutation test would reveal otherwise: a label-dependent step taken before the permutation appears identically in the observed and permuted runs. The property is enforced in one place, `pipeline_core.fold_contained_auc`, which every analysis in this paper calls, and it is checked by an automated gate that fails the build if a `fit_transform` appears outside a training fold in the permutation path.

The **free** test permutes the phenotype vector without restriction; it is `p_free` in the released tables. The **collection-stratified** test permutes it only within collection strata, preserving each stratum's observed class counts; it is `p_collection_preserving` there. A stratum is the interaction of every detected `batch__*` variable with assay, where assay varies. Table 1 names the stratum variables for every comparison, and the screen writes them into its own output (`strata_definition`) so the name and the implementation cannot drift apart. The stratified scheme is the restricted permutation of Chaibub Neto and colleagues [30]; it is applied here, not introduced here.

**The conditioning set is collection variables only.** Sample-quality and demographic columns are in the metadata matrix but not in the strata, so this test conditions on a subset of what is recorded. State the two nulls explicitly. The free test's null is that the phenotype label is exchangeable across all donors; the stratified test's is that it is exchangeable within each collection stratum, that is, conditional independence of expression and phenotype given those strata. Section 3.4 shows what follows: when a sample-quality variable is associated with the phenotype inside a stratum and drives expression, the conditional-independence null is false even though the generating model contains no disease effect, and the test rejects it at rates rising to 95%. Those rejections are correct rejections of a false null, not type-I errors, and they are the reason a significant stratified result does not certify disease biology. Conditioning on the full matrix would need a conditional randomisation scheme for continuous columns, which we do not attempt here.

Where a cohort records no collection variable at all, the fallback is tertiles of log cells per donor. That is a sample-quality variable, so for those comparisons the test is a sample-quality-stratified permutation and is labelled as such rather than being folded in silently.

Where every stratum is label-pure, within-stratum permutation leaves all labels unchanged and the permutation group contains only the observed labelling. The reference distribution is degenerate, and the test is reported as uninformative with p = NA. That is a statement about the permutation group, not a claim that no null distribution exists in any mathematical sense; the free permutation remains defined and is still reported.

A permutation contributes to the collection-stratified null only if at least one stratum contains more than one donor and both labels; if no permutation qualifies, the p-value is undefined and reported as such rather than as a number. Main runs used 1,000 permutations, so the smallest value it can return is $1/1001 \approx 0.0010$. The V_D and metadata-only AUC nulls use 200, so their floor is $1/201 \approx 0.0050$. Every seed used the same counts; the two floors in Table 1 are these two tests, not two different runs.

### 9.7. Feasibility and power checks

A comparison is reported as **not answerable with these data** on one criterion only: no collection stratum holds both classes, so the stratified permutation has no non-trivial reference distribution and no conditioned estimate can be formed. A metadata-only AUC of 1.000 is recorded alongside as corroborating structure, but it is not a second trigger. A finite-sample AUC of 1.000 states that the fitted model separated these donors completely; with 21 donors and seven one-hot columns that can happen without the metadata determining the label in the population, so it is not on its own a proof of non-identifiability. In the one comparison where both hold, the label-pure strata are the stated reason.

A comparison is reported as **underpowered**, and kept in the main ordering with a flag, if it has fewer than 40 donors or fewer than 15 in the minority class. The two gates are separated because they call for different responses: the first cannot be fixed by more careful analysis, and the second can be fixed by more donors.

Restriction has its own feasibility requirement, fixed in advance and applied uniformly: a comparison takes the restricted route only if some collection stratum holds at least five donors of each class and at least 40 donors in total. Where no stratum qualifies the restricted estimate is not computed, and the comparison is reported as needing donors under overlapping conditions rather than being given an estimate from a stratum too small to support one.

The difference between the tuned and the fixed-hyperparameter pipeline AUC is reported as a column (`frozen_minus_tuned_auc`, Table S4) and is deliberately **not** used as a feasibility rule. A gap between two different algorithms does not establish that a comparison is unanswerable. Every inferential statement in this paper is made about the fixed-hyperparameter pipeline, which is the statistic both permutation nulls are built from; the tuned AUC is descriptive. “Fixed-hyperparameter” refers to the statistical pipeline throughout, and “frozen” only to a foundation model whose weights are not updated.

### 9.8. Seed stability

All nine comparisons were re-run at five seeds (20260907-20260911), varying the seed for fold construction, permutation draws, and the cross-fitting split used for V_D and its own permutation null. All five per-seed outputs are released rather than summarised (Figure S7, Table S9), and every range quoted in Table 1 and in the text is the minimum and maximum over those five runs. Five master seeds control fold assignment and permutation draws throughout; the seed is read at call time in every estimator, so each seed reaches the V_D null as well as the classifier.

### 9.9. Simulation

Cohorts of 200 donors and 100 features were generated. A latent variable $z$ produced a site indicator and a continuous quality proxy; the diagnosis was generated as $\rho \cdot z + \sqrt{(1-\rho^2)} \cdot e$ and then either used continuously or dichotomised. Biological and technical loading vectors overlapped partially (technical = 0.7 · random + 0.3 · biological, renormalised). Seven regimes were crossed with $\rho \in \{0, 0.25, 0.5, 0.75, 0.9, 0.98, 1\}$ and both label types, at 200 replicates per cell, giving 19,600 simulated cohorts. The seven regimes are:

1. **additive linear**, the base case;
2. **nonlinear design effect**: tanh plus a quadratic term in the quality proxy;
3. **heteroscedastic noise**: the standard deviation depends on the site;
4. **design-by-cell-type interaction**: two feature blocks whose mixing weight depends on the site, with the biological effect confined to one block;
5. **20% case rate, matched thresholds**: the collection boundary is moved to the same quantile as the diagnosis threshold;
6. **20% case rate, offset boundary**: the collection boundary stays at the median, so the threshold and the boundary are offset;
7. **different acquisition in the target**: the external cohort is acquired under a different design mechanism. The two 20% arms are counted separately because they behave differently, and separating them is what showed the earlier “rare diagnosis blunts the checks” result to be a threshold artefact (Section 3.3).

Critically, the external target reuses the **source’s** biological loading vector. Regenerating it makes every transfer chance-level by construction and tests nothing; only the design mechanism differs between source and target in the shift arm.

A supplementary arm quantifies what admitting a disease-caused covariate does. A severity variable was generated as $1.2 \cdot (y - \bar{y})$ + noise and added to the metadata matrix. Everything else was held fixed.

The extended simulation's cohorts are 200 donors by 100 features, so it applies no feature filter and no PCA; its classifier settings are listed in Table S8 under `extended_simulation` and are deliberately not derived from the screen's, because the screen's 4,000-feature filter and PCA to 50 would be a different reduction on a 100-column matrix rather than the same step. Its unadjusted, residualised and restricted arms differ from each other in exactly one operation, which is the property the calibration study below was missing.

**Calibration arms.** Two further studies use `pipeline_core.SIMULATION`, which *is* the screen's fixed-hyperparameter specification. Arm A places 200 donors in eight collection sites, generates a sample-quality variable that predicts the phenotype within site at strength $\in \{0, 0.3, 0.6, 1.0, 1.5, 2.0\}$, and lets expression depend on the site and on that variable only - there is no phenotype-to-expression arrow. Both permutation tests then run at 200 draws each, over 300 replicates per . Only $= 0$ measures calibration; for $> 0$ the stratified test's conditional-exchangeability null is false by construction, because the sample-quality variable lies outside the conditioning set, so the rejection rate there is the probability of detecting a non-disease association and is reported under that name.

Arm B generates 108 donors and 4,000 features with a real biological effect and a metadata matrix assigned independently of the phenotype, at widths 1 to 92 and in one further layout that reuses the observed level sizes of the CMV batch-by-pool crossing. Unadjusted and residualised AUCs come from the same function with one argument added, so the two arms share folds, feature filter, standardiser, PCA, classifier and scoring rule and differ only in whether the training-fold ridge fit on the metadata matrix is subtracted first. An earlier version fitted the unadjusted arm's PCA on all donors and the residualised arm's inside the fold; the measured gap then contained the change of representation as well as the cost of adjustment, and came out negative at small widths, which is the signature of that asymmetry. Sixty replicates per cell. The sweep is shardable across machines - each cell seeds itself from its own coordinates - and a sharded run reproduces an unsharded one.

### 9.10. Decision-tree walkthrough

Three cohorts were traced through the tree with all evidence recomputed, and every branch value is released in Table S10. The collection strata, the metadata matrix and the classifier are imported from the screen rather than reimplemented, so Q2 and Q4 refer to the same partition, the same columns and the same fitted pipeline; the walkthrough is `SCREEN_FROZEN` at twenty repeats. An earlier version standardised the whole-cohort metadata matrix before handing it to a fold-contained ridge, and ridge shrinkage depends on scale, so a step described as inside the fold was partly outside it; the matrix is now built and standardised on the training donors of each split.

Restriction was evaluated in the largest stratum meeting the feasibility requirement in Section 9.7 - at least five donors of each class and at least 40 in total. Where no stratum qualifies, that is reported and the comparison takes Route C, rather than being worked around by coarsening the strata or by computing an estimate from a stratum too small to support one. Matched random subsets of identical size and case/control composition were drawn 20 times from the full cohort for comparison, which is what separates the cost of conditioning from the cost of using fewer donors.

### 9.11. Recovery of GSE135779 collection metadata

Batch, collection year, age, sex, race, ethnicity and clinical variables were read from the original Supplementary Table 1b, and per-library sequencing metrics from Supplementary Table 1c [1]. Study names were linked to analysis donor identifiers through the GEO title/accession map; all 44 donors mapped uniquely. The restoration script writes the donor table, batch-by-label, year-by-label and batch-by-year cross-tabulations, and SHA256 hashes of every source file. For GSE174188, per-cell `Processing_Cohort` was read from the distributed CELLxGENE object and reduced to donor fractions over four waves. The dominant wave is the maximum-fraction wave; a pure-wave donor has at least 99% of analysed cells in one wave.

### 9.12. Representations

**Frozen Geneformer.** Cell embeddings used Geneformer V2-316M at repository revision `04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5` [4]; checkpoint, configuration, token dictionary and gene-median dictionary hashes are in the methods manifest. Gene identifiers were version-trimmed and mapped to the V2 vocabulary. Within each cell the ranking value was `raw count / total count × 10,000 / Genecorpus-104M gene median`, and genes were sorted descending. Sequences used V2 start/end token IDs 2 and 3, padding ID 0 and maximum length 4096, leaving at most 4094 ranked genes. The model output was the final `last_hidden_state`; token position 0 was kept as the 1,152-dimensional cell vector. The encoder was frozen. Donor representations are coordinate-wise means of sampled cell vectors, sampled without replacement with integer seed 1, capped at 500 cells per donor for GSE135779 and 1,000 for GSE174188 and GSE285773. The embedding environment used Python 3.10, PyTorch 2.5.1 and Transformers 4.46.3.

**Pseudobulk.** We use pseudobulk rather than a learned latent space as the donor-level representation, so that the comparison between representations is not itself mediated by a model fitted with batch covariates [5]. Stored donor-by-gene values are `log1p(CPM)`. Inside every outer training fold, gene variance was computed on training donors, the 4,000 highest-variance genes were selected, and coordinates were standardised by training-donor mean and standard deviation. HVG pseudobulk uses these values directly. PCA pseudobulk fits up to 30 components on the training values and applies the fitted map to held-out donors. For cross-cohort transfer, trimmed gene identifiers were intersected before source-only HVG selection; feature means, variances, selected genes, scaler, PCA, classifier and regularisation were all fitted on source donors and applied unchanged to the target.

### 9.13. Fold-contained residualisation and its variants

Unadjusted and residualised models share outer folds, feature construction, scaling, PCA and classifier selection. Inside each outer training fold, metadata variables were encoded as in Section 9.5 and a multivariate ridge regression (`alpha` 1, intercept fitted and **not** penalised) mapped design to every raw representation coordinate. Training and held-out coordinates were replaced by observed minus predicted values, after which the identical pipeline was fitted to training residuals and applied to held-out residuals. The diagnosis was never given to the residualiser.

Three further arms were run on GSE135779. A multivariate random-forest residualiser (96 trees, depth 4, minimum leaf 3) fitted only on outer-training donors, with pseudobulk genes pre-selected by training-donor variance. A batch-location arm subtracting training-batch mean offsets from training and held-out features; because it applies no empirical-Bayes scale shrinkage we describe it as ComBat-style location adjustment rather than ComBat. And an overlap-weighting arm fitting a training-only logistic diagnosis propensity model from the full design, clipping propensities to 0.025-0.975 and training the representation classifier with overlap weights.

For the central GSE174188 comparison, residualisation loss was paired by split seed. We bootstrapped the 20 paired repeat-level losses 5,000 times and subtracted the locked matched-restriction discrepancy. These intervals quantify split sensitivity conditional on the analysed donors; they are not population confidence intervals.

To probe confound leakage with a nonlinear downstream learner [28], 100 Gaussian simulations used no biological effect, a technical effect of one, and metadata-phenotype association 0.75. Random-forest prediction was compared on raw features, globally residualised features and fold-contained residualised features.

### 9.14. Restriction and matched donor controls

GSE174188 was restricted to dominant wave 4 and separately to pure wave 4. GSE135779 excluded the all-case batch B1 and retained B2 to B6, each containing both labels. The complete internal pipeline was re-run after restriction. For every observed stratum, 20 donor subsets were drawn without replacement from the full cohort with identical donor count and case/control count, each using one prespecified split seed. The observed-minus-matched difference asks whether the stratum is harder than expected from reduced donor number and label composition alone.

### 9.15. Cell-type composition

The GSE174188 object was restricted to the same CD4-positive alpha-beta T-cell population and 261 donors as the molecular analysis. Per-donor counts were formed for `T4_naive`, `T4_em`, `T4_reg` and all other or unassigned labels. We evaluated raw closed proportions and centred-log-ratio coordinates; for CLR, 0.5 was added to each donor-category count before closure, the donor mean log abundance was subtracted and the final coordinate omitted. Each representation was evaluated with and without log total CD4-cell yield, under the same repeated donor pipeline, with processing wave predicted one against the rest.

### 9.16. Transfer and learned pooling

Cross-batch analyses trained on waves 2 and 3 and evaluated wave 4, then reversed. Cross-cohort transfer fitted every statistic on the source cohort; target labels were accessed only after prediction. Target AUC intervals used 5,000 case/control-stratified donor bootstrap resamples and the percentile method. Paired AUC comparisons used DeLong tests on identical target donors [23] with Benjamini-Hochberg correction inside declared method families [24].

DeepSets, gated-attention multiple-instance learning and pooling by multi-head attention operated on cap-500 frozen cell embeddings [20-22]. Cell standardisation, a whitened source PCA32 projection, network weights and hyperparameters were all source-fitted. Hidden widths were 16 or 32, weight decay 0.01 or 0.1, and Adam used learning rate 0.02 for 150 epochs; five-fold source-donor cross-validation selected the configuration, three initialisations were fitted on all source donors, and target probabilities were averaged. An ordinary donor mean in the identical PCA32 space isolates the contribution of learned pooling. The PCA32 projection used for hyperparameter selection was fitted once on all source cells before source-donor folds and never used target cells or labels. Source-internal pooling AUC is therefore not an unbiased generalisation estimate, and only independent-target and paired target-donor comparisons support the learned-pooling conclusion.

### 9.17. Software and reproducibility

The donor-level analyses used Python 3.11-3.13 with NumPy 2.3-2.5, pandas 2.3-3.0, scikit-learn 1.9.0, PyArrow and joblib; exact per-run versions are recorded in each output manifest. Figures were produced in R 4.6.0 with ggplot2 4.0.3 and patchwork 1.3.2. The screen, the extended simulation, the permutation calibration and the decision-tree walkthrough were run on a 16-vCPU ARM container; all other analyses were run locally. Input hashes, seeds, parameter grids, donor-level predictions, complete null distributions and output manifests accompany the analysis.

Random seeds are derived with `hashlib.blake2b` from a recorded master seed and the cell identity. Python's built-in `hash` is not used: it is salted per process for strings, so it would not reproduce across interpreter sessions. The derived seed for every cell is written into the released result tables. No script contains a machine-specific path: the figure scripts locate the project root from their own file position or from `RHEUMLENS_ROOT`, and the analysis scripts take input and output paths as arguments.

Everything downstream of the compute-heavy runs is rebuilt by one command, `bash tools/build_all.sh`: the supplementary tables, Table 1, all fifteen figures, the typeset manuscript, and two checks that fail with a non-zero exit status. The first re-derives numbers quoted in the text from the result tables and requires each to appear verbatim. The second checks the length of the abstract and author summary, that the abstract is unstructured, that figures and tables are cited in the order they are declared, and that each declared item is cited at least once. It also refuses any placeholder, retired term or withdrawn phrase, and it looks inside the rendered figures and the released table headers, where a check on the manuscript text alone cannot see them. The submission package is assembled by a further script from the same outputs, so a figure number cannot differ between the manuscript and the files.

### 9.18. Use of generative AI

Generative AI coding assistants were used during this work: Anthropic Claude (models Claude Opus 4.1 and Claude Opus 5) through the Claude Code command-line interface, and OpenAI Codex through its command-line interface. They were used in the following places.

- **Code review and refactoring** of the analysis scripts in `scripts/cohorts/`, `sim/` and `walkthrough/`. Every statistical definition, gate and threshold in those files was specified by the authors; the model's contribution was implementation, review of implementation, and the identification of two defects that the authors then confirmed by inspection and by rerunning the affected analyses.
- **Figure code.** All figures were drawn by R scripts written with model assistance (`figures/src/*.R`). Every number plotted was read from a released result table, and the correspondence between figure and table was checked by the authors panel by panel.
- **Manuscript language and structure**: rewriting for length and readability, reorganisation of sections, and consistency checking of cross-references and reference numbering.

No text, number, figure or citation was accepted without author verification against the underlying result tables. Cohort selection, the statistical definitions, every gate and threshold, the decision of which analyses to report, and the interpretation of results were made by the authors. No data of any kind were generated by a model: every value in this paper traces to a released result table produced by the analysis scripts. The authors take full responsibility for the content of this publication.

## 10. Data and code availability

All cohorts are public. The lupus datasets are at GEO accessions GSE135779, GSE174188 and GSE285773; the COVID-19, influenza and CMV cohorts were obtained through the CELLxGENE Discover curation API and their dataset identifiers are in `cohorts/registry.yaml`. The MIT-licensed repository is at https://github.com/LightChainr/rheumlens with a permanent Zenodo record at https://doi.org/10.5281/zenodo.20813922. The versioned research object accompanying this manuscript contains the cohort registry with API-verified donor counts, the donor-level interface tables, every result table underlying Figures 2 to 8, all five per-seed outputs rather than a summary, the 19,600-cohort simulation output, analysis scripts, locked environments, `REPRODUCE.md` and `SHA256SUMS`. Public raw single-cell matrices remain at their original accessions.

## 11. Declarations

**Author contributions.** Conceptualization, H.Y. and D.L.; methodology, H.Y.; software, H.Y.; validation, H.Y. and D.Y.; formal analysis, H.Y.; data curation, H.Y.; writing - original draft preparation, H.Y.; writing - review and editing, D.Y. and D.L.; visualization, H.Y.; supervision, D.L.; project administration, D.L. All authors have read and agreed to the submitted version of the manuscript.

**Funding.** This research received no external funding.

**Competing interests.** The authors declare no competing interests.

**Ethics approval.** Not applicable. This study analysed only publicly available, de-identified single-cell datasets released by their original investigators under the consent and approvals obtained at each contributing site. No new human or animal data were collected.

**Informed consent.** Not applicable, for the reason given above.

**Use of AI tools.** See Section 9.18, which states what was used, where, and how the output was checked.

**Acknowledgments.** The authors thank the investigators of GSE135779, GSE174188 and GSE285773, and the contributors to the COMBAT, Stephenson, Ren and CMV cohorts dis-

tributed through CELLxGENE Discover, for releasing donor-level data that made this re-analysis possible.

## Figure legends

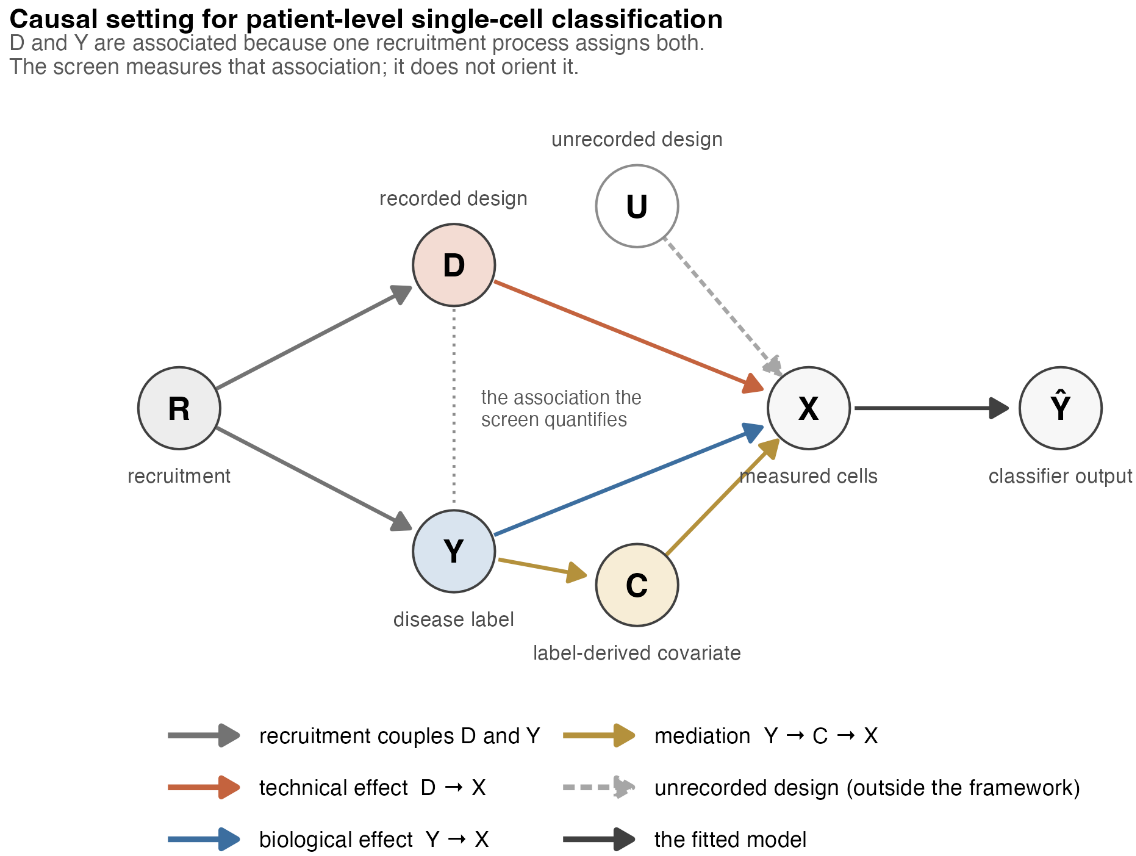


**Figure 1. What is being measured.** A recruitment process R places each donor in the recorded design D and is also associated with the diagnosis Y; the measured cells X are affected by D (technical) and by Y (biological); a classifier maps X to $\hat{Y}$. C denotes covariates caused by the disease - severity, medication, symptom timing - which lie on $Y \to C \to X$ and are excluded from D by blocklist. U denotes collection structure that was never recorded and which nothing in this paper can see. The dotted line marks the association these checks quantify. They measure it; they do not tell you which way it runs.

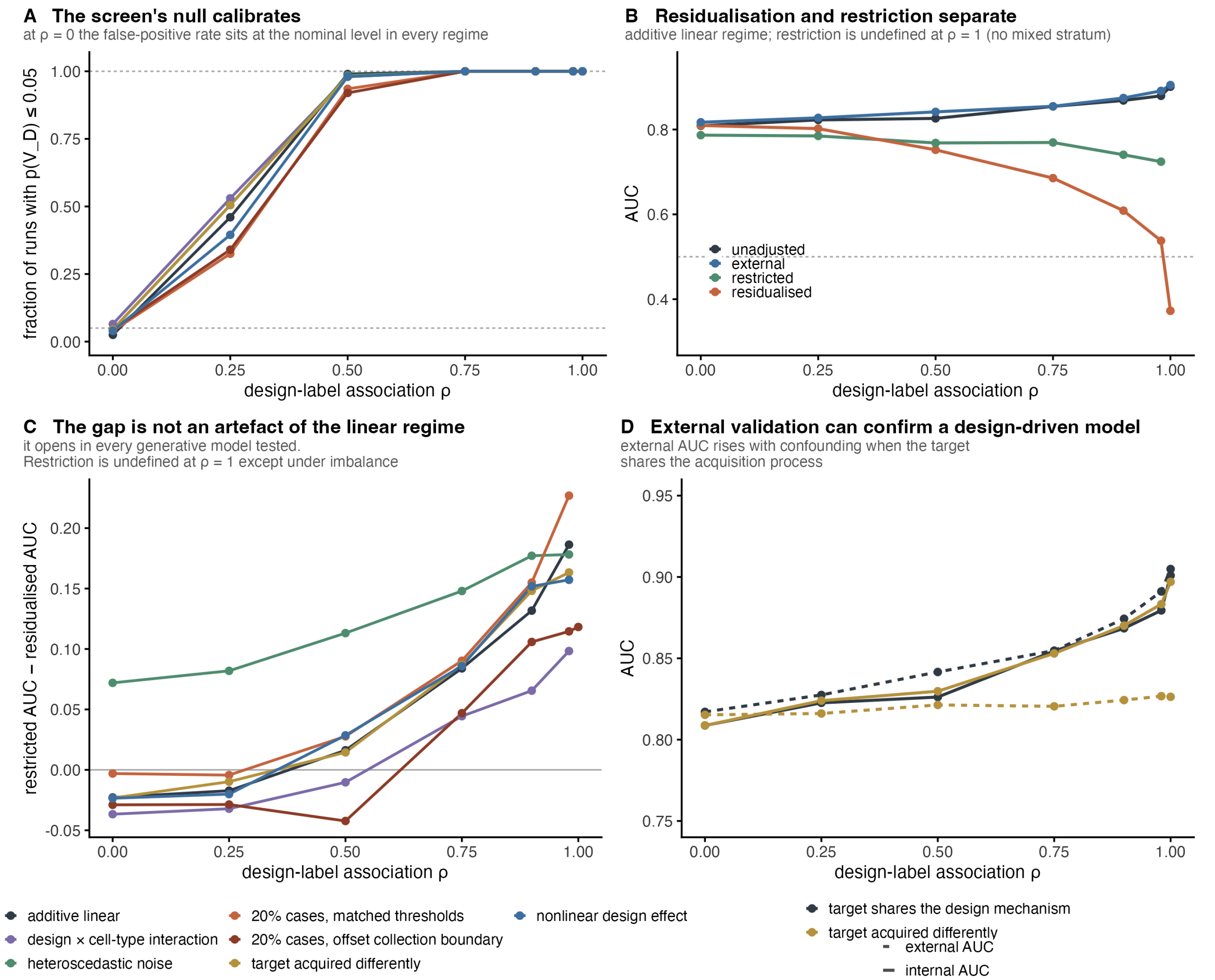


**Figure 2. Simulation across seven data-generating regimes.** 19,600 simulated cohorts (200 donors, 100 features): seven regimes × seven levels of metadata-phenotype association $\rho$ × two label types × 200 replicates. **(A)** Fraction of runs with $p(V\_D) \leq 0.05$, binary labels. At $\rho = 0$ this is 0.025 to 0.065 across regimes, bracketing the nominal 0.05. **(B)** Unadjusted, external, restricted and residualised AUC against $\rho$ in the additive linear regime. Restriction is undefined at $\rho = 1$ because no stratum holds both labels. **(C)** Restricted minus residualised AUC for all seven regimes. The gap opens in every one. It is blunted in the `imbalanced_offset` regime, where the collection boundary and the phenotype threshold sit at different quantiles, and not in `imbalanced`, where the case rate is the same 20% but the two are matched (Section 3.3). **(D)** Internal (solid) and external (dashed) AUC when the target shares the source's collection mechanism versus when it does not. External AUC rises with confounding in the shared case.

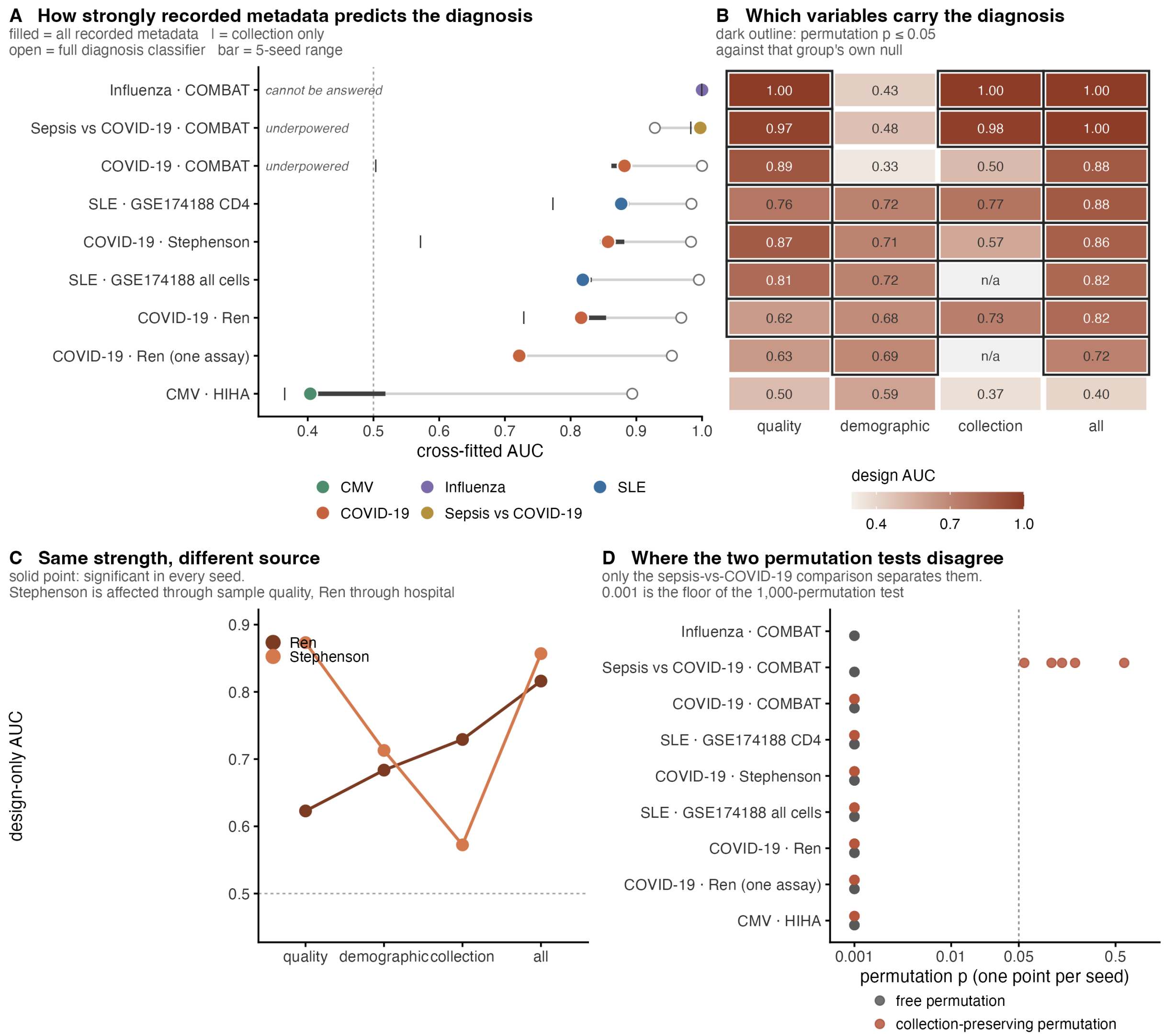


**Figure 3. Nine case-control comparisons. (A)** Comparisons ordered by cross-validated metadata-only AUC (filled, coloured by disease) with the corresponding diagnosis-classifier AUC (open) on the same axis. Bars give the range over five random seeds. Dashed line is chance. Italic labels mark the two comparisons that the feasibility check removes from or flags within the ordering. **(B)** Metadata-only AUC by variable group, rows aligned to (A). A dark outline marks $p \leq 0.05$ for the metadata-only AUC against that group's own permutation null; "n/a" marks a group absent for that comparison. **(C)** The two COVID-19 cohorts with almost equal overall metadata-only AUC are affected through different variable groups: Stephenson through sample quality, Ren through recruiting hospital. Solid points are significant. **(D)** Free and collection-stratified permutation p-values, one point per seed, for all nine comparisons. Dashed line is 0.05. Only the

sepsis-versus-COVID-19 comparison separates the two tests.

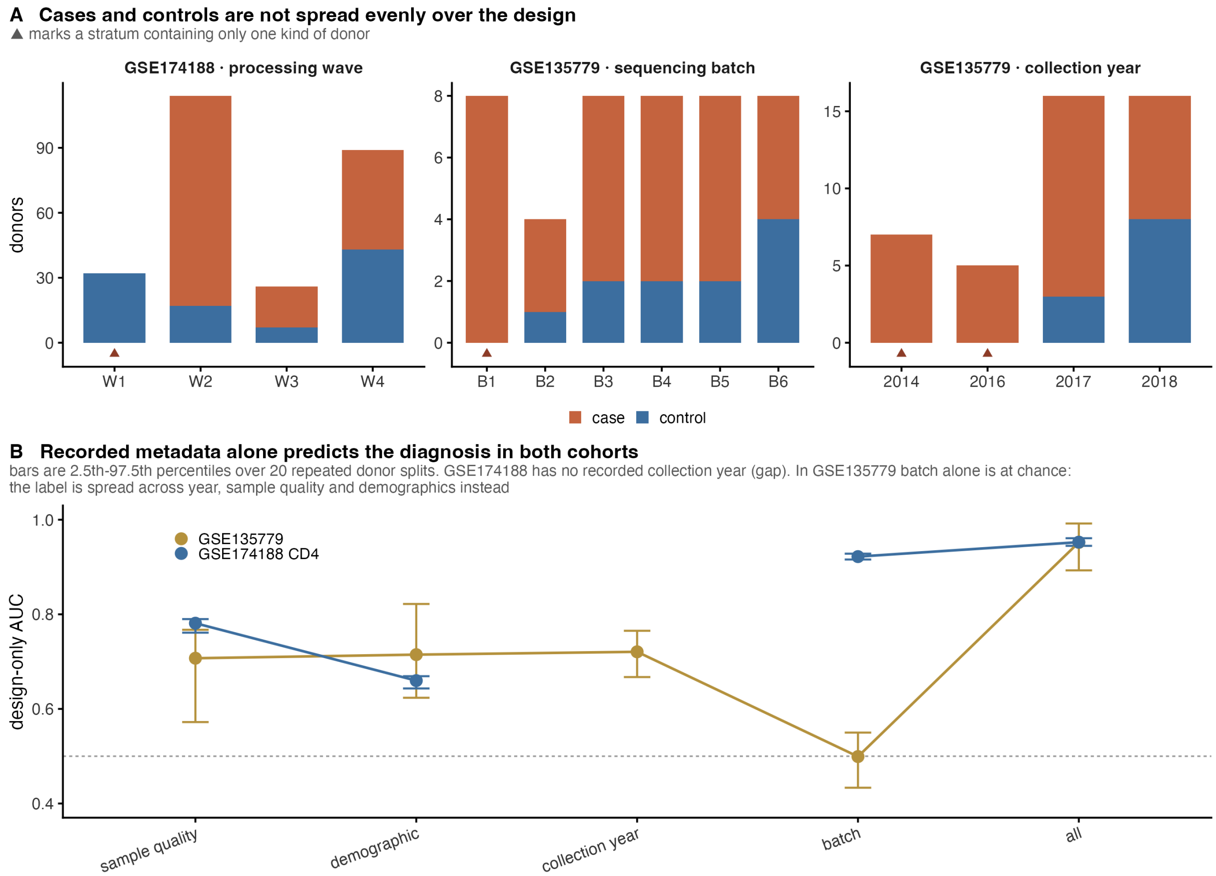


**Figure 4. Design and diagnosis in the two lupus cohorts. (A)** Donors per collection stratum, split by case and control, for processing wave in GSE174188 and for sequencing batch and collection year in GSE135779. A triangle marks a stratum holding only one kind of donor; such strata cannot be permuted within, and cannot be restricted to. **(B)** Cross-validated AUC for predicting the diagnosis from recorded metadata alone, by variable group, with 2.5th-97.5th percentiles over 20 repeated donor splits. GSE174188 has no recorded collection year, so its line is broken there. In GSE135779 batch alone sits at chance (0.499) while the other groups do not, so the association runs through collection year, sample quality and demographics rather than through batch.

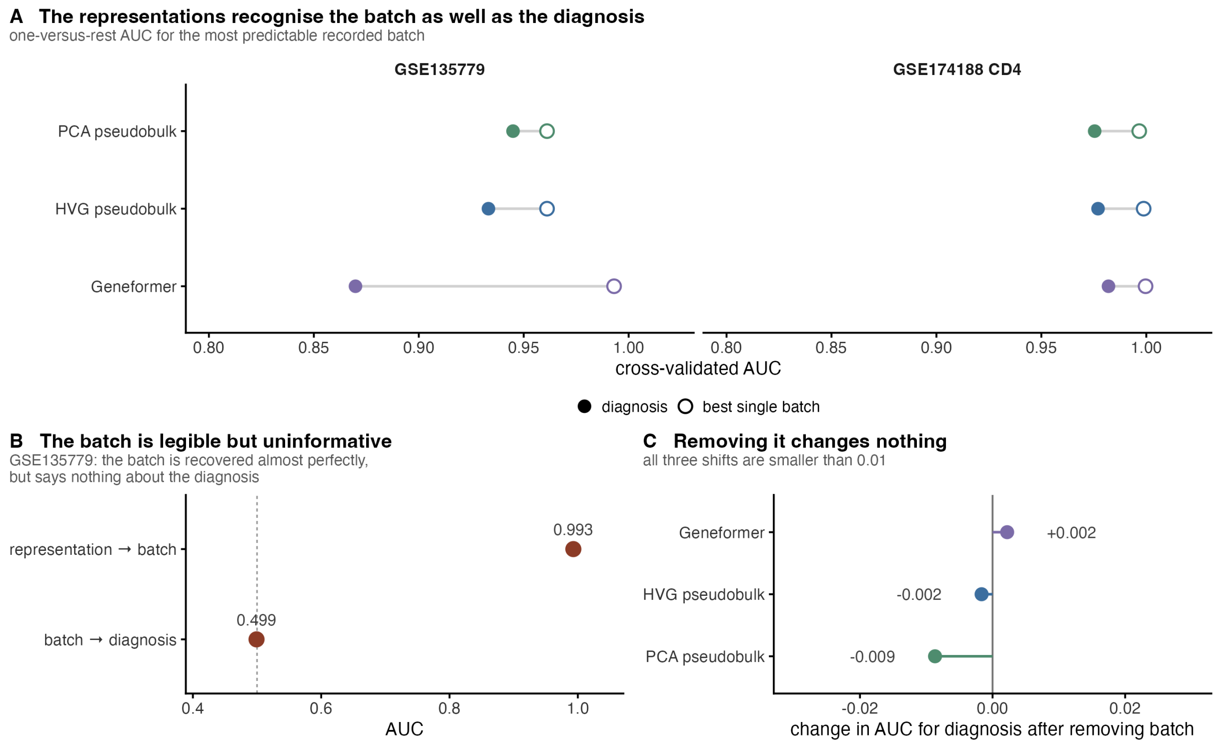


**Figure 5. Containing batch information is not the same as using it. (A)** For each representation, the cross-validated AUC for the diagnosis (filled) and for the most predictable single recorded batch, one against the rest (open), in both cohorts. Every representation recognises a batch about as well as it recognises the diagnosis. **(B)** GSE135779 negative control. The representation recovers the batch at AUC 0.993, but the batch predicts the diagnosis at 0.499. **(C)** Change in disease AUC after fold-contained batch residualisation in the same cohort: +0.002, −0.002 and −0.009, all inside the split-to-split spread. Batch is predictable from the representation, unpredictive of disease, and its removal changes little, so batch predictability on its own supports no conclusion about what the classifier uses.

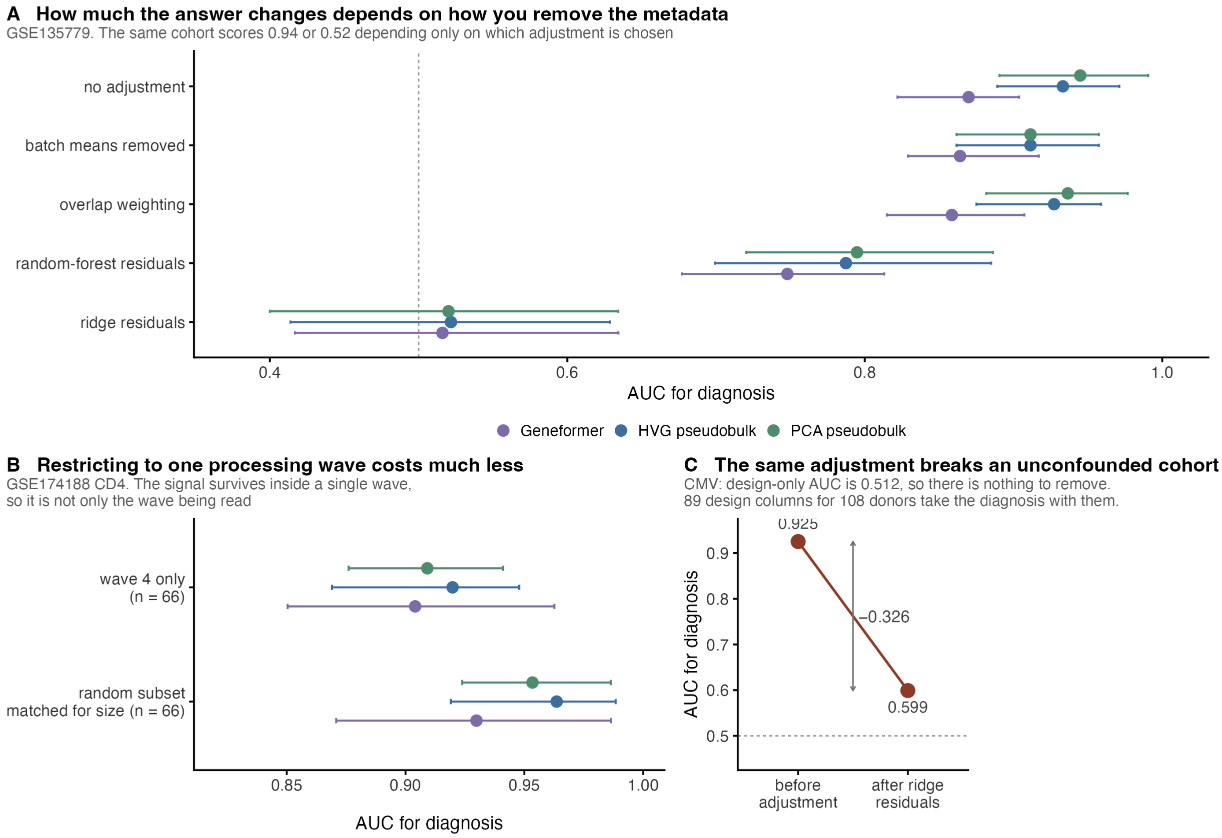


**Figure 6. Residualisation and restriction give different answers, and neither is identified.** **(A)** GSE135779 diagnosis AUC under five ways of removing the recorded metadata, for three representations, with 2.5th-97.5th percentiles over 20 repeated splits. The same cohort scores 0.94 or 0.52 depending only on which adjustment is chosen. **(B)** GSE174188 CD4, restricted to the one large near-balanced processing wave ($n = 66$), against size-matched random donor subsets. The separation largely survives inside a single wave. **(C)** Negative control. In the CMV comparison, where the metadata-only AUC is 0.512 and no metadata-phenotype association is detectable in any seed, the same ridge residualisation still costs 0.326 AUC. A simulated matrix of the same ragged shape, with the phenotype independent of the metadata by construction, costs 0.363 on average (central 95% 0.272-0.464; Figure S2). Residualisation loss is a function of the shape of the adjustment matrix and cannot be read as a measure of contamination.

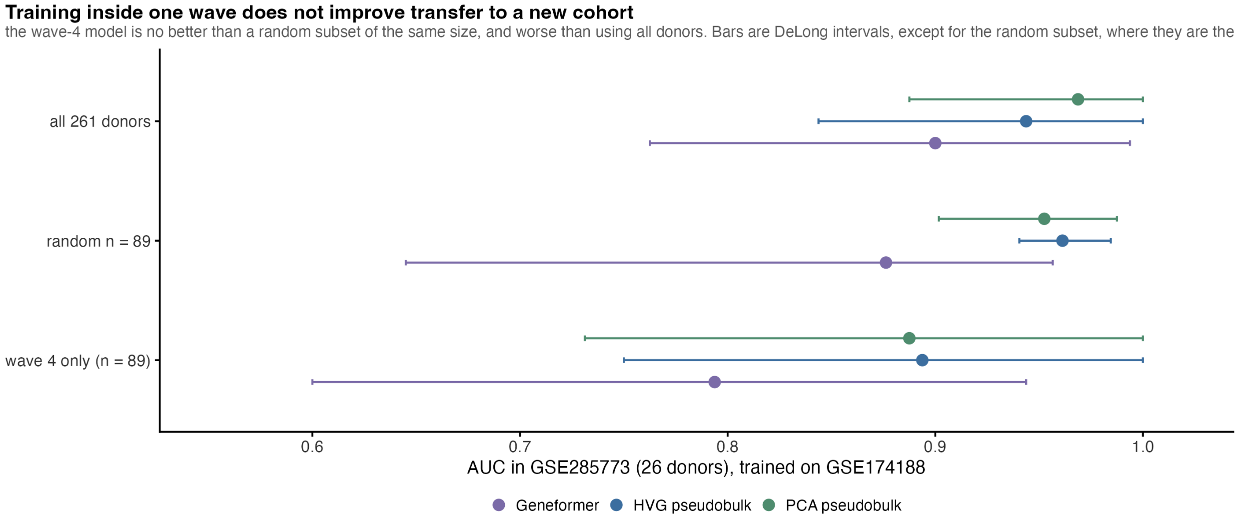


**Figure 7. Restricting the training set to one processing wave does not improve transfer.** AUC in GSE285773 (26 donors) for classifiers trained on GSE174188, by source set: all 261 donors, a random subset of 89, and the dominant processing wave (n = 89). The wave-restricted source is no better than a random subset of the same size and worse than using every donor. Bars are DeLong intervals, except for the random subset, where they are 2.5th-97.5th percentiles over 20 draws.

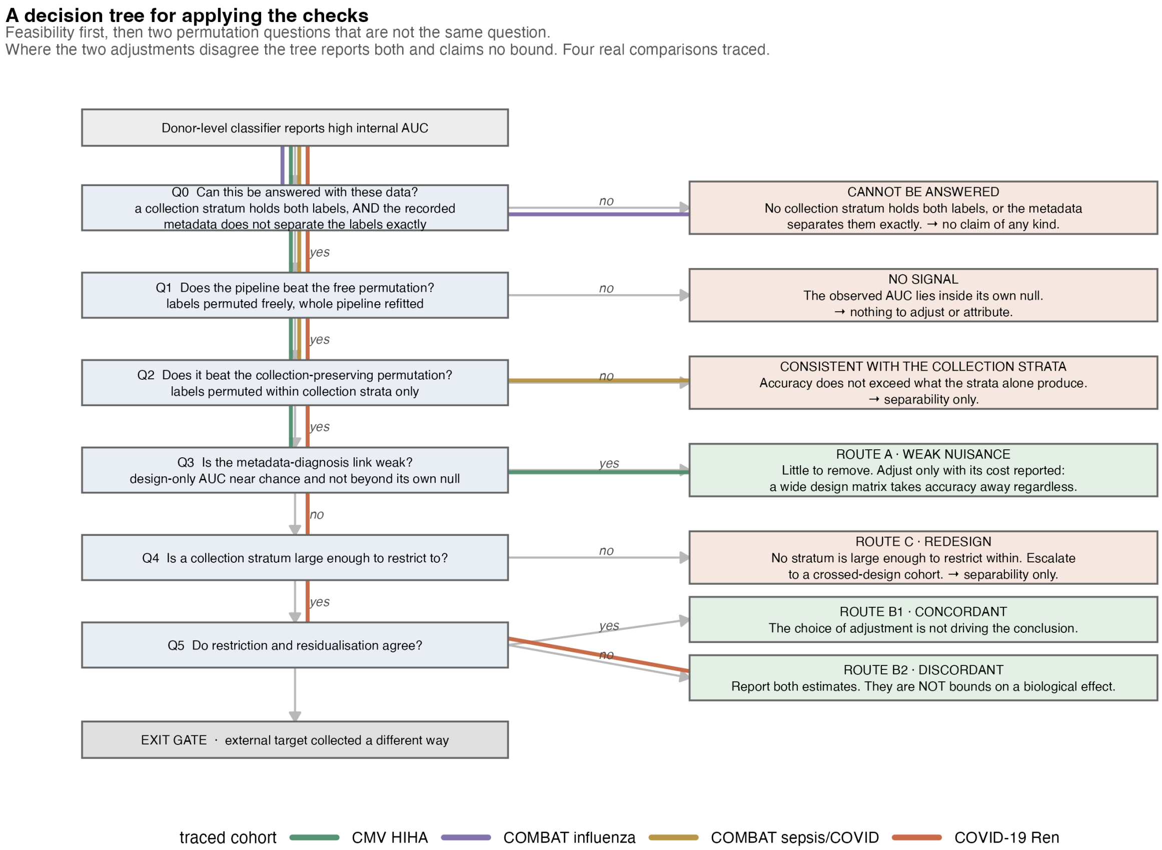


**Figure 8. A decision tree for applying the checks.** The feasibility question comes first (Q0): a comparison is set aside when no collection stratum holds both labels, so that no conditioned estimate can be formed. The permutation step is then split into two questions that state different nulls - the free test (Q1) and the collection-stratified test (Q2). Q3 reads the metadata-only AUC against its own permutation null rather than against a fixed cut-point; a null result there reports that no collection association was detected and leaves the unadjusted estimate standing, and does not recommend an adjustment. Q4 requires restriction to be feasible - some stratum with at least five donors of each class and at least 40 in total - so each comparison has one output; where it is not, the comparison takes Route C. Every terminal box states what can be estimated and what is still missing, not a remedy to apply. Coloured paths trace four real comparisons. The strata are the ones the collection-stratified permutation uses, not a separate definition.

## Supporting information

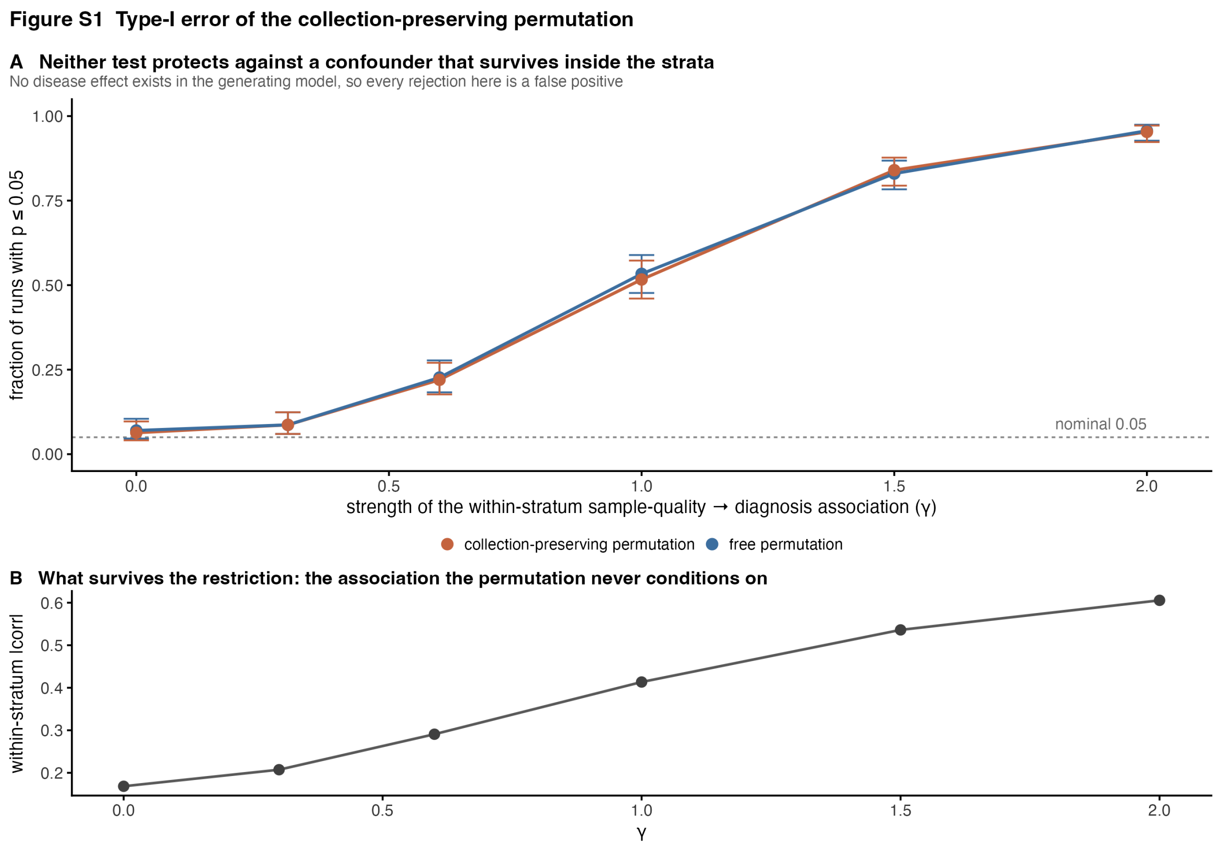


**Figure S1. Rejection rate of both permutation tests under a non-disease within-stratum association. (A)** Fraction of runs rejecting at $p \leq 0.05$ for both permutation tests, as a function of how strongly a sample-quality variable predicts the phenotype *within* a collection stratum. The generating model contains no disease effect at all. Only the leftmost point ( = 0) measures calibration; to its right the conditional-exchangeability null is false by construction, because the sample-quality variable lies outside the conditioning set, so the rate is the probability of detecting a non-disease association rather than a type-I error rate. **(B)** The mean absolute within-stratum correlation that conditioning on the collection strata leaves in place.

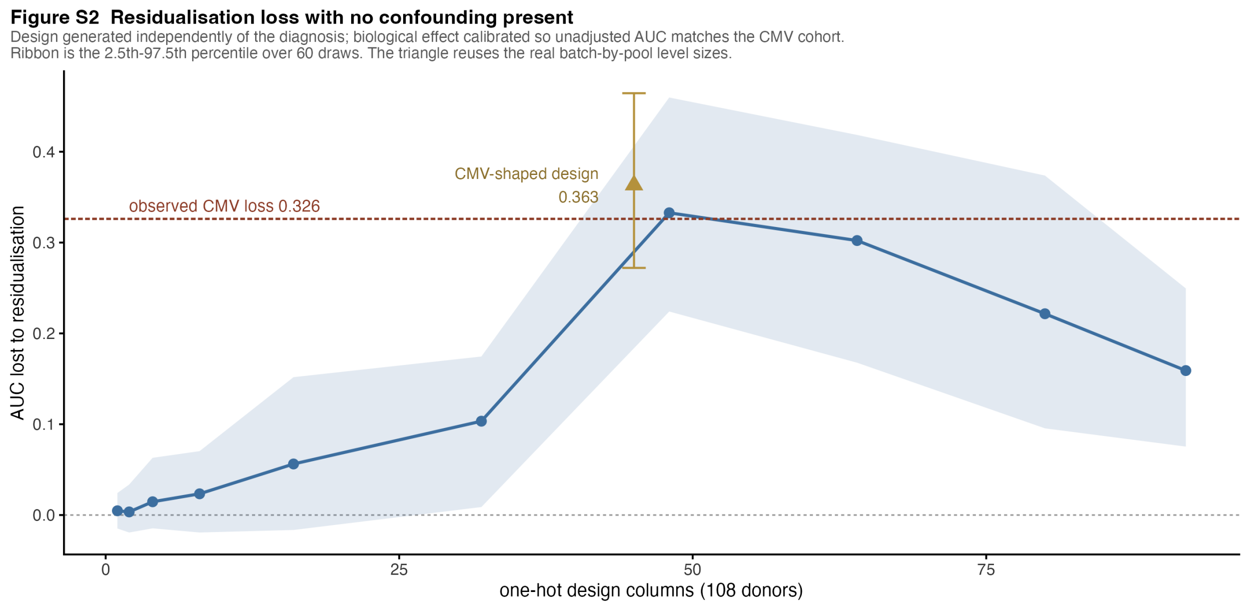


**Figure S2. Residualisation loss with no metadata-phenotype association present.** AUC lost to fold-contained ridge residualisation in simulated cohorts where the metadata matrix is independent of the phenotype by construction, against the width of that matrix, at 108 donors. The unadjusted and residualised arms are the same call with one argument added, so they share folds, feature filter, standardiser, PCA and classifier and differ only in the adjustment. The biological effect is calibrated so the unadjusted AUC lands near 0.90. The ribbon is the 2.5th-97.5th percentile over 60 draws. The triangle reuses the level sizes of the real CMV batch-by-pool crossing, which is far more ragged than equal blocks; it loses 0.363 on average. The dashed line marks the 0.326 observed in that comparison, which falls inside the triangle's interval.

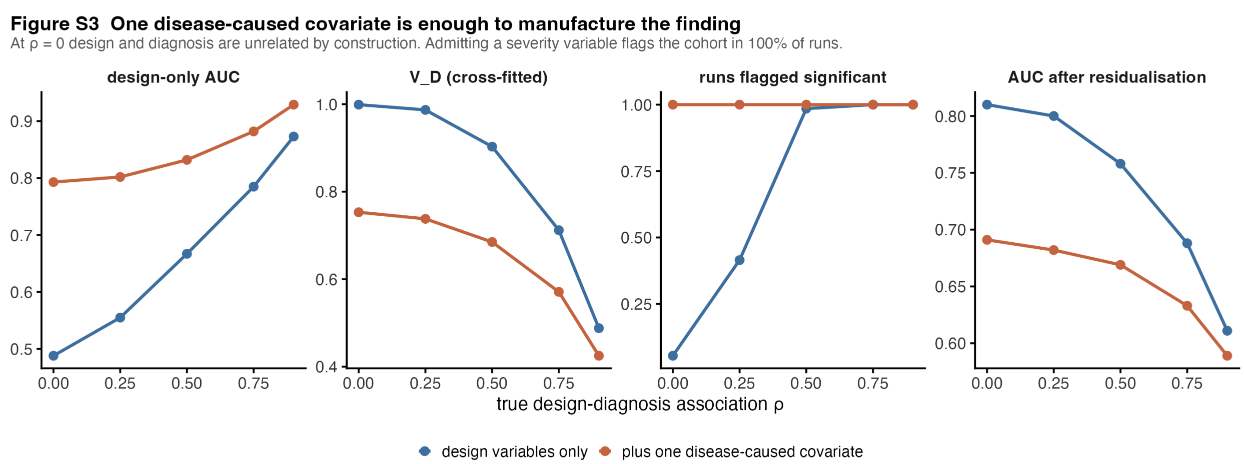

**Figure S3. Admitting one disease-caused covariate manufactures the finding.** Metadata-only AUC, V_D, the fraction of runs flagged significant, and residualisation loss, with and without a severity variable added to the metadata matrix, across five levels of true metadata-phenotype association. At $\rho = 0$, where metadata and phenotype are unrelated by construction, admitting the covariate raises metadata-only AUC from 0.488 to 0.793 and flags the cohort in 100% of runs against 5.5% without it.

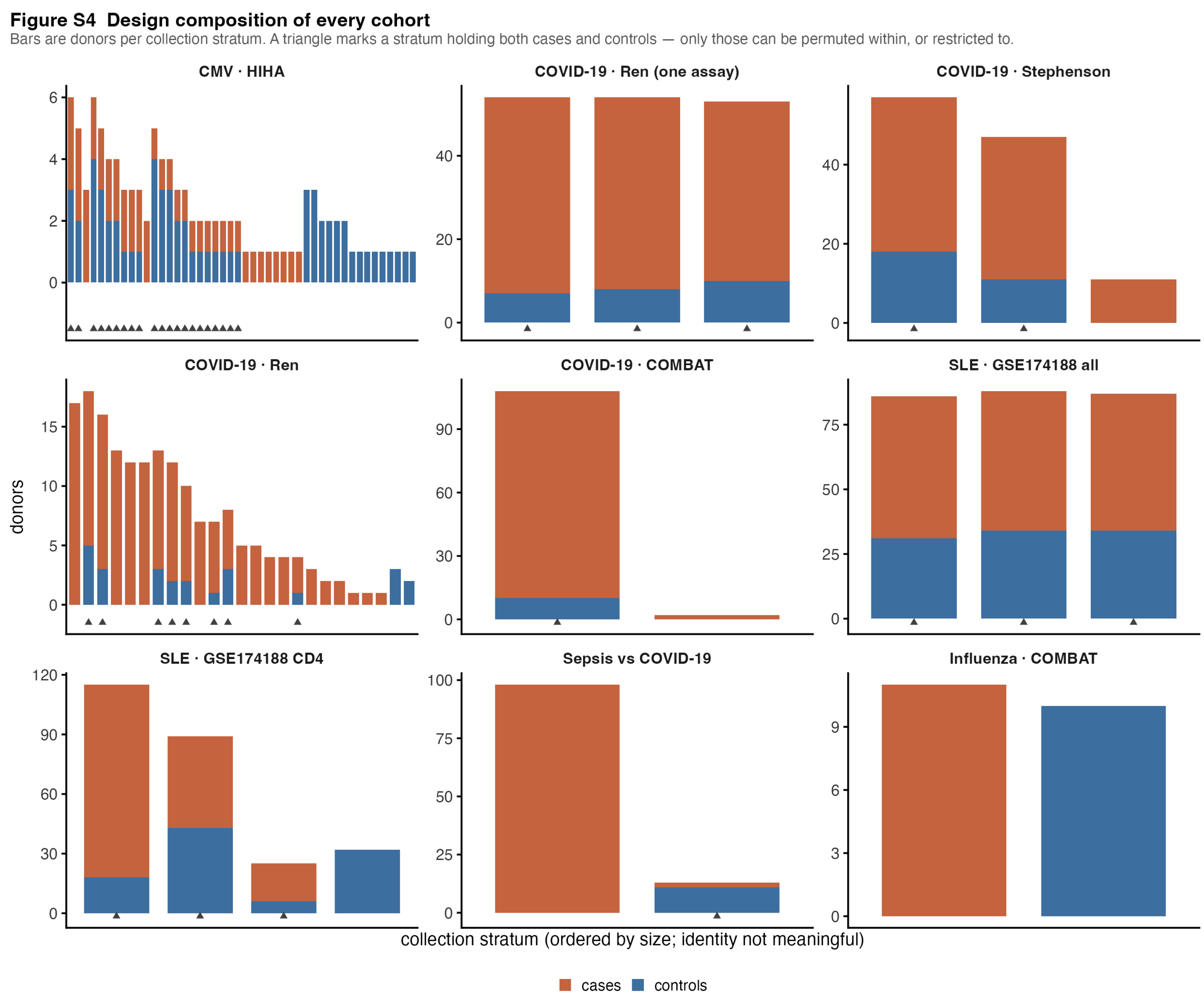


**Figure S4. Design composition of each cohort.** Case and control counts by collection stratum for all nine comparisons, showing which strata are label-pure and how many donors sit in mixed strata.

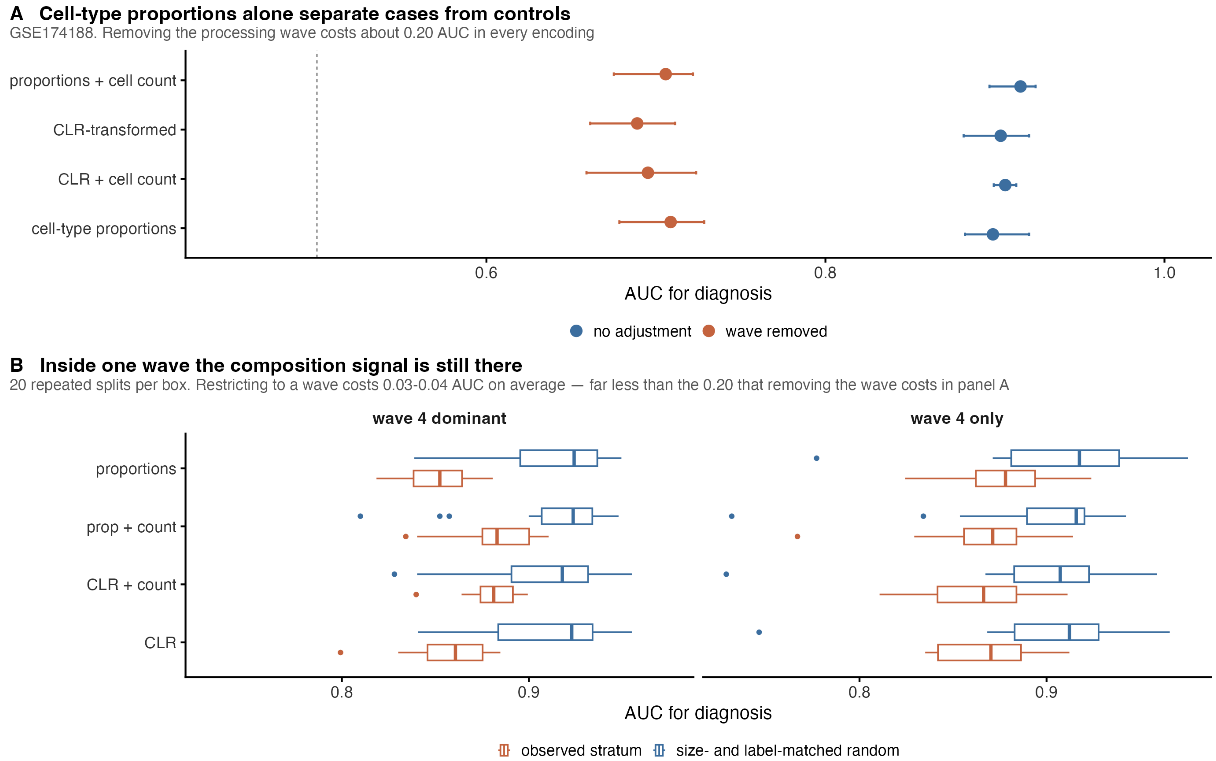


**Figure S5. Cell-type composition reproduces the whole pattern. (A)** Diagnosis AUC in GSE174188 from four cell-type fractions alone, under four encodings, before and after removing the processing wave. Removing the wave costs about 0.20 AUC in every encoding. **(B)** The same encodings restricted to the dominant wave and to the pure wave, each against a size- and label-matched random donor subset, 20 repeated splits per box. Restriction costs 0.03 to 0.04 AUC on average - far less than removal costs in (A). A four-number biological summary shows the same behaviour as a 1152-dimensional foundation-model embedding, which places the problem in the cohort rather than in the embedding.

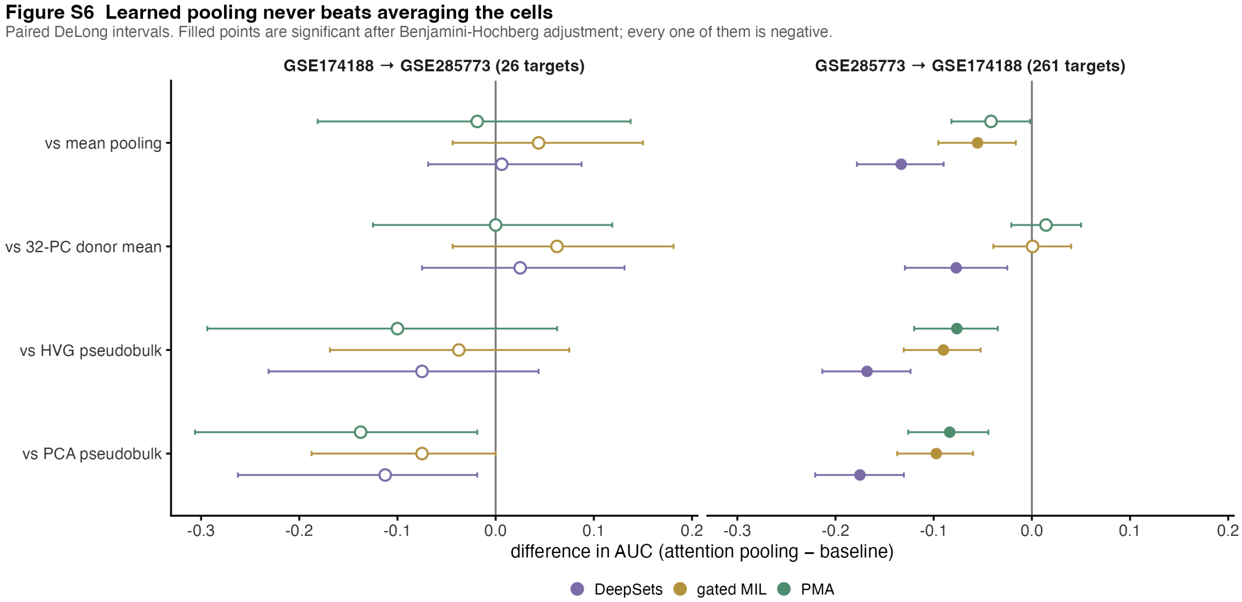


**Figure S6. Learned pooling never beats averaging the cells.** Paired differences in AUC between three learned pooling methods (DeepSets, gated-attention multiple-instance learning, pooling by multi-head attention) and four baselines, in both transfer directions, with paired DeLong intervals. Filled points are significant after Benjamini-Hochberg adjustment; every significant difference is negative.

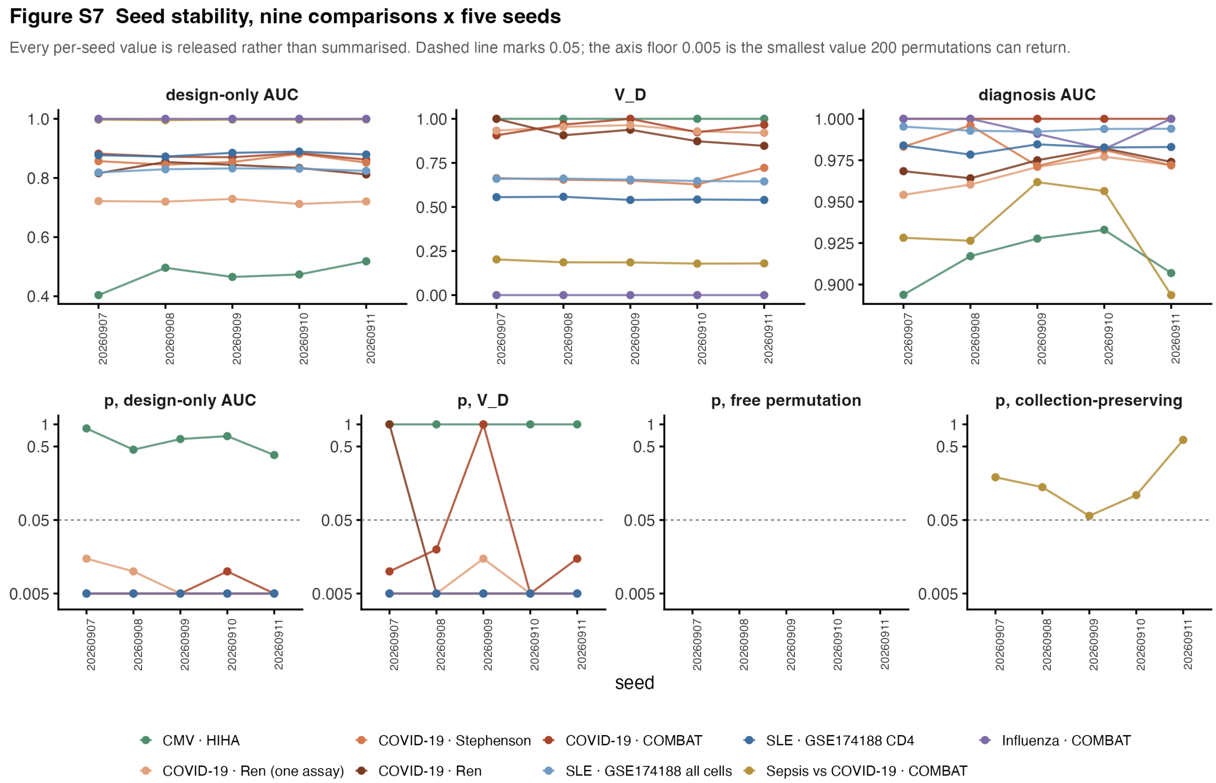


**Figure S7. Seed stability.** Per-seed values of metadata-only AUC and its permutation p, V_D and p(V_D), diagnosis AUC and both permutation p-values, for all nine comparisons at five seeds. The p-value panels use a log axis; 0.005 is the smallest value 200 permutations can return.

**Table S1.** Calibration study: summary output of both arms, the permutation rejection-rate error of Section 3.4 and the residualisation-width study of Section 3.5. **Table S2.** Design manifest, generated directly from the model-matrix construction code: for every comparison and every variable group, the raw columns consumed and the number of expanded columns produced. **Table S2b** lists every column present in a covariate file that enters no block, with the reason. **Table S2c** is a machine check that the manifest reproduces the width of every fitted model matrix (34 of 34 blocks agree). **Table S3.** Simulation summary: all seven regimes × seven $\rho$ × two label types, plus the disease-caused-covariate arm. **Table S4.** Full screen output: every comparison × every variable group, with metadata-only AUC (linear and random forest), V_D in-sample and cross-fitted, null mean, p(V_D), diagnosis AUC, both permutation p-values and stratum counts. **Table S5.** Metadata-only, expression-only and joint AUC for every comparison at each of five split seeds, with the two increments. All three arms use the same fixed-hyperparameter specification and the same folds; expression is reduced to principal components fitted on the training donors before the metadata columns are appended. **Table S6.** Learned pooling: per-method target metrics and all paired DeLong tests, both transfer directions. **Table S7.** Cohort registry, generated from the registry file and the donor tables: for all seven datasets, the identifier, citation, the donor count reported by the source, the download size, the case and control labels, the donor, case and control counts actually modelled in each comparison, and how and when each entry was verified. **Table S8.** Every fitted pipeline in the paper and its settings, generated from the specification objects in `pipeline_core.py` that the analysis scripts import. The four analyses are separate pipelines and the table keeps them separate. **Table S9.** Seed-stability table, all five seeds, all nine comparisons. **Table S10.** Decision-tree walkthrough: all evidence for every branch, three cohorts.

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